

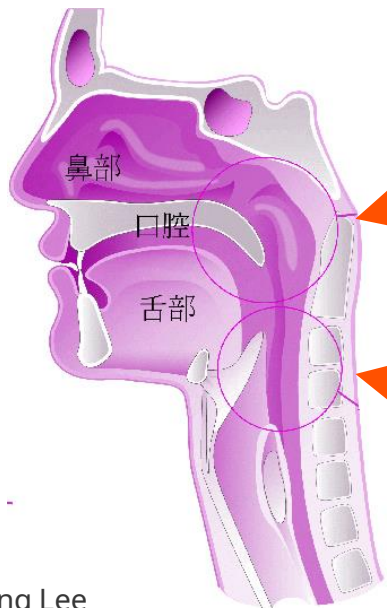
流感防治 與疫苗接種政策

主講人：李秉穎 醫師
臺大兒童醫院



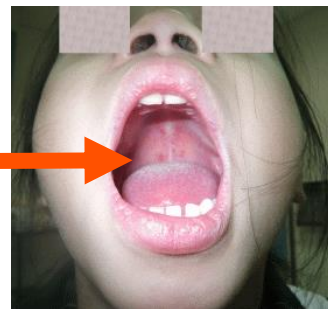
流感不等於感冒或上呼吸道感染

Influenza $\neq$ common cold, URI



感冒

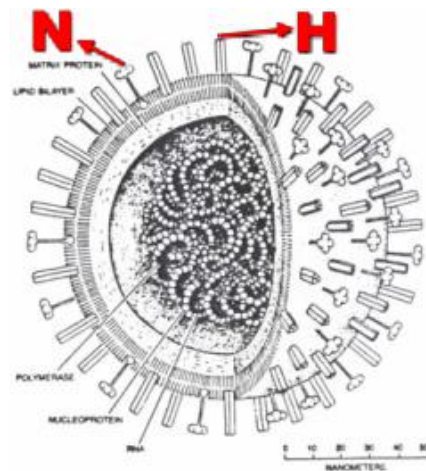
流感



Photo's taken by Dr. Ping-Ing Lee

流感 = Influenza

- A、B二型為人類重要傳染病
- 抗原：近幾十年流行株H3N2, H1N1
 - Hemagglutinin (H, 血凝集素)
 - 鳥類H1-H18，人類 H1-H3
 - Neuraminidase (N, 神經胺酸酵素)
 - 鳥類N1-N11，人類N1-N2

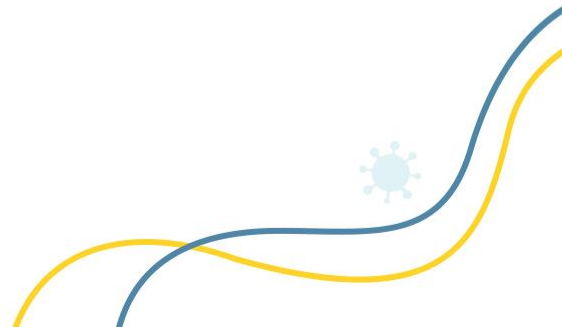


流感 = Influenza



■ 突變種類

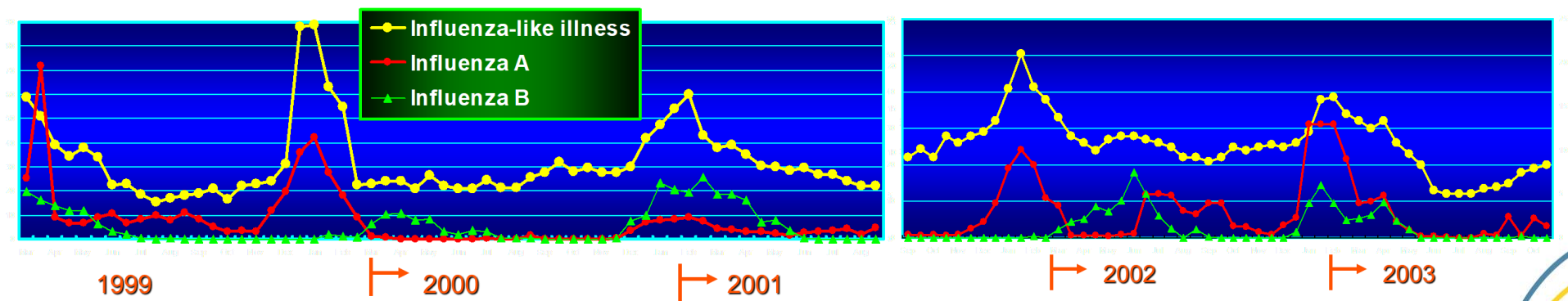
- **Antigenic drift**，小突變：依照發現地命名，例如雪梨型、北京型、福建型等
- **Antigenic shift**，大突變：引起全世界大流行 (pandemic)



臺灣流感病毒監測



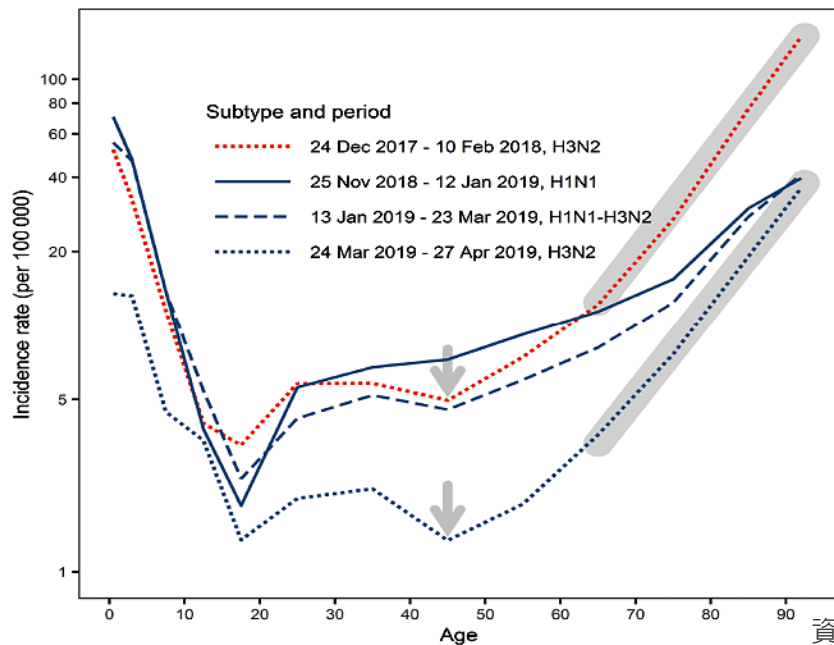
Average reported cases per physician (% positive)



資料來源：Taiwan CDC

Age-specific incidence of influenza

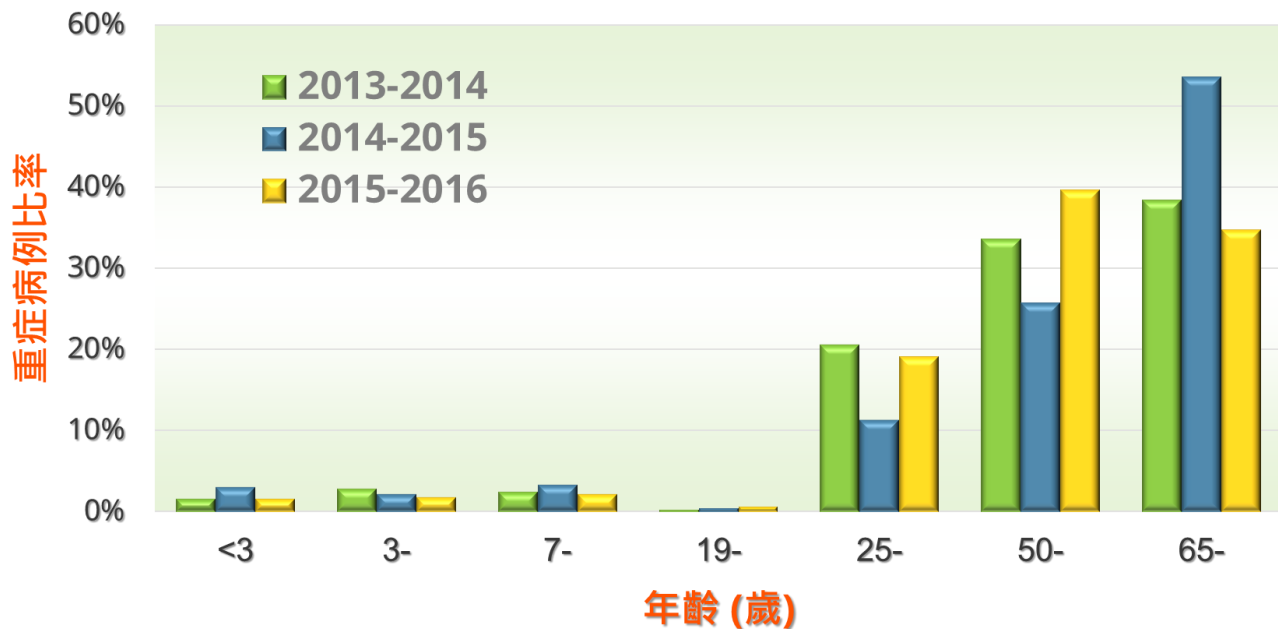
2017~2018, Quebec



資料來源：Gagnon A. Clin Infect Dis 2020;71(7):e195.

2013-2016年臺灣流感併發重症 年齡分布

2013.7.1 - 2016.4.30，疾病管制署流感速訊



抗流感藥物



■ 神經胺酸酶抑制劑 (neuraminidase inhibitors) : A型與B型流感均有效

- Zanamivir (Relenza, 瑞樂沙) :
 - 吸入型藥物, 2 puff bid X 5 days
- Oseltamivir (Tamiflu, 克流感) : 口服
 - 劑量 : 75 mg bid X 5 天
 - 副作用 : 腸胃不適 (約10%)
- Peramivir (Rapiacta) : 靜脈注射

■ 抑制RNA polymerase inhibitor

- Favipiravir (Avigan) : 口服5天, 孕婦與兒童不能使用

■ cap-dependent endonuclease inhibitor

- Baloxavir : 口服一劑



流感之治療建議

臺灣疾病管制署



- 不一定需要使用抗流感藥物：非屬重症高風險族群或高傳播族群
- 建議使用抗流感藥物
 - 高風險族群
 - 高傳播族群
 - 病程快速進展，出現危險病徵者
 - 無危險徵兆之健康兒童，若希望縮短病程，可考慮給予治療。

資料來源：疾病管制署(2022)。流感防治與疫苗接種政策 公版簡報。



公費流感抗病毒藥劑使用對象一覽表

適用日期：114年8月1日起

除法定傳染病外，應為本國籍人士或持有居留證
【18歲(含)以下孩童其父母需一方為本國籍或持有居留證】治療性用藥條件 經醫師評估符合條件者，無須快篩即可用藥

法定傳染病

需通報於法定傳染病通報系統及填寫法傳編號

- 「流感併發重症」通報病例
- 「新型A型流感」通報病例

住院病患

確診或疑似罹患流感住院(含急診待床)之病患

罹患流感因病況嚴重而需住院治療的病患，並不包括門診病人，依此條件使用公費藥劑者須備有「住院紀錄」或「急診待床紀錄」

門急診病患

- 未滿5歲及65歲以上之類流感患者
- 孕婦及產後兩週內之婦女經評估需及時用藥者
領有國民健康署核發孕婦健康手冊之婦女
- 肥胖之類流感患者(BMI $\geq$ 30或未滿18歲兒童青少年BMI超過同齡第95百分位)
未滿18歲兒童青少年BMI第95百分位一覽表詳如附件
- 具重大傷病、免疫不全(含使用免疫抑制劑者)或流感高風險慢性疾病之類流感患者
 - 重大傷病：IC卡註記為重大傷病或持有重大傷病證明紙卡者
 - 流感高風險慢性疾病之ICD CODE：B20, Z21, D80-84, D86, D89, E08-13, E66, E85, G09, G20, G30-32, G35-37, G40, G45-46, G65, G70, G72, I00-02, I05-09, I11-13, I20-22, I24-25, I27-28, I34-37, I42-43, I44-45, I47-49, I50-51, I60-62, I63, I67-69, I70, I71, I72, I73-74, I77, I79, J40-45, J47, J60-70, J82, J84, J96, J98, J99, K70-72, K73-76, B18-19, M05-06, M30-31, M32-34, M35, M94.1, N00-01, N03, N05, N04, N18-19, N26-27, Q89.01, Z90.81

預防性用藥條件

需通報並經審核

需通報衛生局進行疫情調查，並經疾病管制署各區管制中心防疫醫師或傳染病防治醫療網區正/副指揮官或其授權人員同意後始可用藥

- 類流感等群聚事件經疾病管制署各區管制中心防疫醫師認定需用藥者
需填寫群聚編號
- 「新型A型流感」極可能/確定病例之密切接觸者
接觸者名冊經傳染病防治醫療網區正/副指揮官或其授權人員研判需給藥者;需填寫所接觸之個案的法傳編號
- 動物流感發生場所撲殺清場工作人員
接觸者名冊經傳染病防治醫療網區正/副指揮官或其授權人員研判需給藥者；需填寫禽畜場名稱或編號

未滿 18 歲兒童青少年 BMI 第 95 百分位一覽表

附件

以兒童及青少年生長身體質量指數 (BMI) 建議值(衛福部 102 年 6 月 11 日公布)肥胖為依據。

年紀	男性	女性
0.0	15.8	15.5
0.5	19.9	19.6
1.0	19.2	19.0
1.5	18.5	18.2
2.0	18.3	18.1
2.5	18.0	17.9
3.0	17.8	17.8
3.5	17.7	17.8
4.0	17.6	17.9
4.5	17.6	18.0
5.0	17.7	18.1
5.5	18.0	18.3
6.0	18.5	18.8
6.5	19.2	19.2
7.0	20.3	19.6
7.5	21.2	20.3
8.0	21.6	20.7
8.5	22.0	21.0

年紀	男性	女性
9.0	22.3	21.3
9.5	22.5	21.6
10	22.7	22.0
10.5	22.9	22.3
11	23.2	22.7
11.5	23.5	23.1
12	23.9	23.5
12.5	24.2	23.9
13	24.5	24.3
13.5	24.8	24.6
14	25.0	24.9
14.5	25.2	25.1
15	25.4	25.2
15.5	25.5	25.3
16	25.6	25.3
16.5	25.6	25.3
17	25.6	25.3
17.5	25.6	25.3

流感群突發事件處理原則



- 每日監測服務對象及工作人員（含流動工作人員）之健康狀況
- 如出現發燒應依「人口密集機構傳染病監視作業注意事項」進行通報
- 依據傳染病及流行病學原理，確定病例定義，選定調查範圍與方法，採集人員或環境檢體並收集相關資料輔助分析調查
- 應限制感染區域與非感染區域人員的跨區移動
- 配合主管機關進行接觸者追蹤、疫情監測或預防性投藥等防治工作，以降低疫情擴散

資料來源：長期照護機構季節性流感感染管制措施指引 (疾病管制署2017.10.13)



流感群突發的預防性投藥



- Oseltamivir或zanamivir一天一次，一共10天
 - 由衛生局向疾病管制署各區管制中心提出預防性投藥申請
 - 預防性投藥原則：
 - 確認為流感陽性(含快篩結果)之事件
 - 儘速於暴露後48小時內投藥
 - 以「感染流感後易併發重症之高危險族群」為原則
 - 曾與發病個案密切接觸者之高危險住民
 - 曾密切照護或接觸發病個案者之流感高危險醫護與工作人員優先使用瑞樂沙
 - 持續投藥至與該次群聚事件最後一例發病個案接觸後10天
- 
- 

流感群突發的預防性投藥

- 流感高危險族群：(1)65歲以上老人；(2)孕婦；(3)未滿5歲幼兒；(4)具有重大傷病、免疫不全、慢性肺病(含氣喘)、心血管疾病、腎臟、肝臟、神經、血液或代謝疾病者(含糖尿病)、血紅素病變等需長期治療、追蹤或住院者；(5)肥胖(身體質量指數 ≥ 30)

非活化流感疫苗



■ 非活化疫苗：傳統疫苗在雞胚培養

- 三價疫苗：A(H1N1) + A(H3N2) + B
- 四價疫苗：A(H1N1) + A(H3N2) + B (Victoria) + B (Yamagata)

■ 保護效果：

- 保護效力: 40%-60%
- 每年必須接種一劑

■ 副作用：

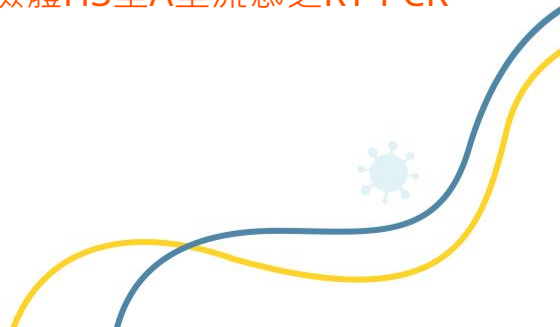
- 注射部位腫痛：成人>兒童
 - 發燒：幾乎不會
- 

流感疫苗

每年均需接種一次



- 62歲女性
- 過去病史：高血壓、糖尿病
- 2004年接種流感疫苗，2005年因身體不舒服未接種疫苗
- 2005.3.17 發燒、腹瀉
- 2005.3.20 北市某醫院急診處就醫後返家
 - 當天嚴重虛弱且無法下床，住院
- 2005.3.21 呼吸困難、血氧下降轉加護病房
 - 胸部X光呈成人呼吸窘迫症候群變化
 - 開始給予Tamiflu 1# bid
 - 晚上6:50因為休克死亡
 - 呼吸道檢體H3型A型流感之RT-PCR陽性



流感疫苗的保護持續時間

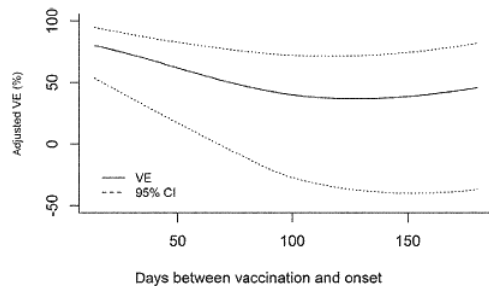
2011~2015, USA



- Decline in vaccine effectiveness was more pronounced among patients with prior-season influenza vaccination.

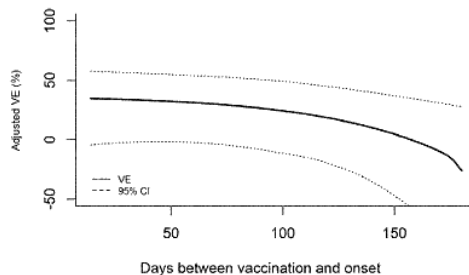
A(H1N1): effectiveness

↓ 6-11%/month



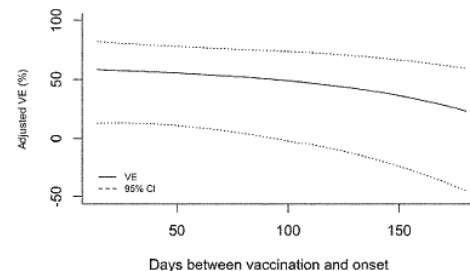
A(H3N2): effectiveness

↓ 7%/month



B: effectiveness

↓ 7%/month



資料來源：Ferdinands JM. Clin Infect Dis 2017;64(5):544.

流感疫苗的保護持續時間

2015~2019, CDC, USA



- Decline in VE against hospitalization: 8%/month, faster in the elderly

Table 1. Estimated Decline in Influenza Vaccine Effectiveness per Month Postvaccination Among Adults Enrolled in the United States Hospitalized Influenza Vaccination Network (HAIVEN), 2015–2016 Through 2018–2019

Influenza Type/Subtype	Influenza Seasons Included	No. of Cases/Controls	Estimated VE Decline per Month, Absolute % (95% CI)	P Value ^a
Influenza A(H3N2)^b				
Aged ≥18 y	2016–2017, 2017–2018	754/2262	7.5 (.3–16.3)	.05
Aged ≥65 y	2016–2017, 2017–2018	395/1185	10.8 (2.6–23.8)	.02
Influenza A(H1N1)pdm09^c				
Aged ≥18 y	2015–2016, 2018–2019	373/1119	8.5 (3.0–17.0)	.003
Aged ≥65 y	2015–2016, 2018–2019	132/396	9.6 (–3.3 to 32.7)	.14
Influenza B/Yamagata^b				
Aged ≥18 y	2016–2017, 2017–2018	265/795	8.0 (1.4–21.9)	.02
Aged ≥65 y	2016–2017, 2017–2018	134/402	10.8 (1.4–33.9)	.03

資料來源：Ferdinands JM. Clin Infect Dis 2021;73(4):726.

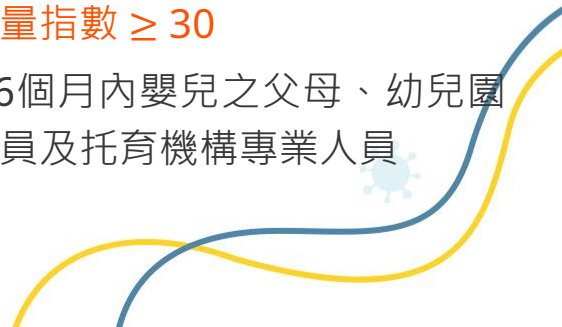


臺灣流感疫苗免費接種對象 之政策演進

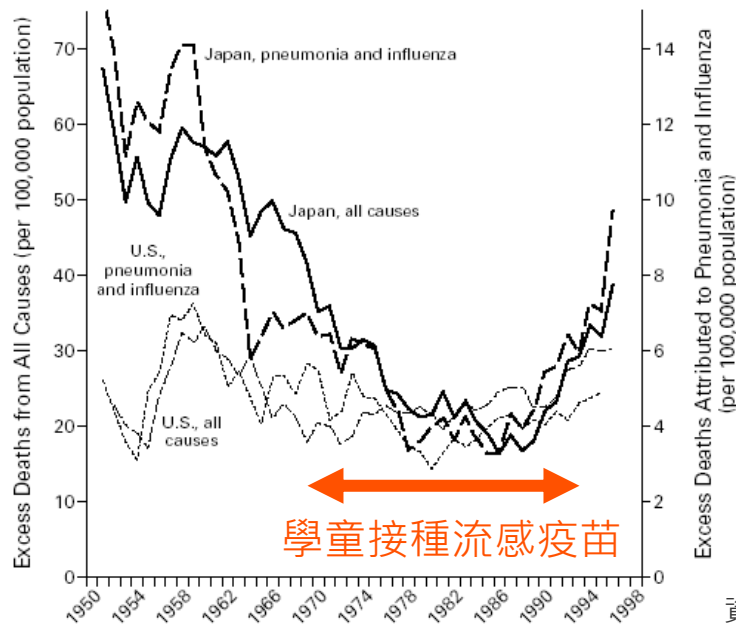


※紅字為高風險族群

- 2001 65歲以上老人
- 2003 醫護工作人員、衛生單位防疫相關人員、禽畜養殖相關人員
- 2004 6個月至2歲兒童
- 2007 國小1-2年級學生、衛生保健志工
- 2008 2-3歲幼兒、國小3-4年級學生、重大傷病患者
- 2009 3歲至國小入學前幼兒、空中救護勤務人員
- 2012 國小5-6年級學生
- 2013 60-64歲具高風險慢性病患
- 2014 孕婦、50-59歲具高風險慢性病患
- 2016 國中、高中/職及五專1-3年級學生、產後6個月內婦女、50-64歲健康成人、未滿50歲高風險病患、身體質量指數 ≥ 30
- 2017 6個月內嬰兒之父母、幼兒園托育人員及托育機構專業人員



學齡兒童：高傳播族群



日本

1957 8000死亡

1962學童接種：7-15歲
接種率 50-85%

1977強制接種

1987自由接種

1994停止

資料來源：Reichert TA. New Engl J Med 2001;344:889.

流感與運動

- 盧彥勳敗給感冒 (2008.2.7 民生報)：盧彥勳受感冒影響，因體力不支，決定忍痛退賽
- HBL / 關毓萱吐怨氣 北一女逆轉勝 (2011.1.5 聯合報)：趁著普門全隊被感冒侵襲，北一女以58：54逆轉取勝
- 梅西重感冒致使出現呼吸困難 巴薩隊醫全程陪護 (2012.9.6 中國網)
- 擋不住艾卓吉 火箭難堪2連敗 (2014.4.25 中國時報)：帶著感冒未癒身體，林書豪替補上場24分43秒
- 小約重感冒 墨瑞不戰而勝 (2015.1.4 聯合晚報)：墨瑞兵不血刃抱走冠軍

流感疫苗對健康成人的保護效果

美國1994-1995, 18-64歲, N=849

■ 流感疫苗組

- 上呼吸道感染次數 ↓ 25%
- 呼吸道疾病病假天數 ↓ 43%
- 因上呼吸道感染就醫次數 ↓ 44%

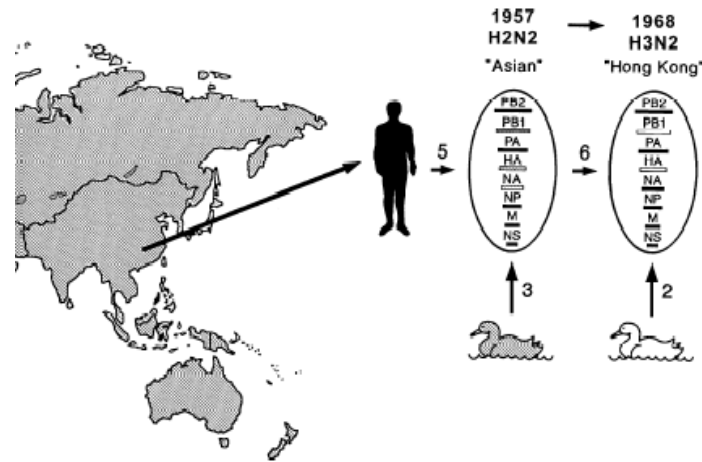
■ 節省成本：\$46.85/person

■ 美國從2011年開始建議所有6個月以上兒童與成人，每年都接種一劑流感疫苗

資料來源：Nichol KL. N Engl J Med 1995;333(14):889.

國外建議的疫苗在臺灣 是否有效？

- Epidemic strain vs. vaccine strain
 - Taiwan, 1997-2004: match rate
H3N2 82%, H1N1 53%, B 47%¹
- China as an epicenter²



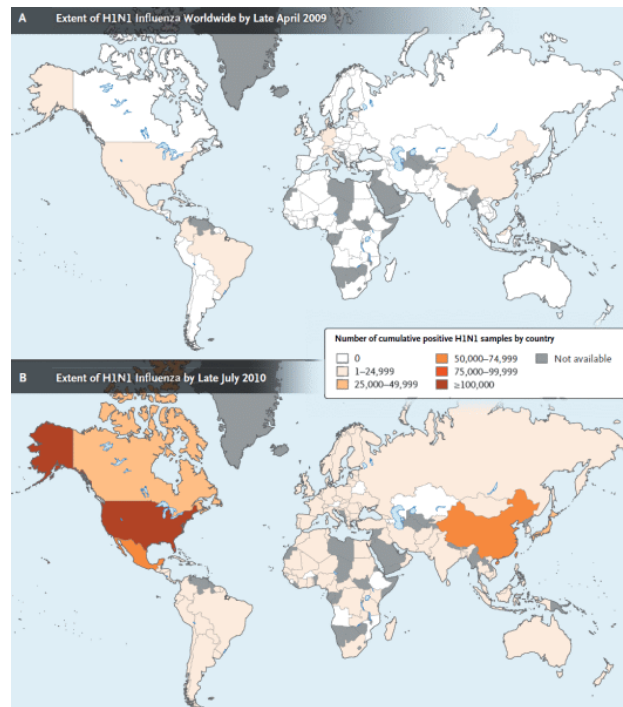
1. Hsieh YC. J Microbiol Immunol Infect 2005;38:238.
2. Webster RG. Vaccine 2003 (Suppl 2):16.

H1N1 新型流感的快速傳播

2009年4月底



2010年7月



資料來源：Fineberg HV. N Engl J Med 2014;370:1335.

公費流感疫苗猜錯病毒株

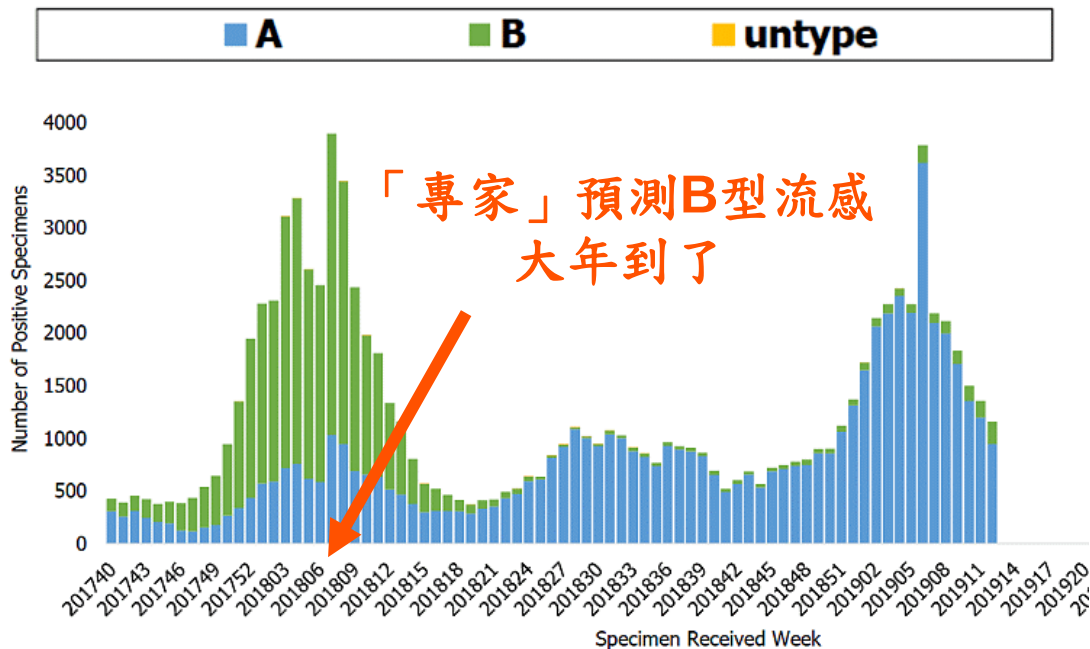
流感「大年」到了

2018.1.2 聯合報 記者李樹人

- XXX以流感「大年」來形容這次流感疫情，主要原因為今年世界衛生組織猜錯了B型流感病毒株，原本預測為維多利亞型，但實際上流行的是日本山形株，...三分之二感染了不在疫苗保護範圍內的山形株，造成多名病童因罹患腦炎、肺炎而住院。
- XXX說，從去年11月起，就驚覺不對勁，臺大兒童門診的類流感患者、重症患者明顯增加，12月病例數越來越多，目前已有多名幼兒因為感染B型山形株，而引發腦炎、肺炎，住進加護病房。

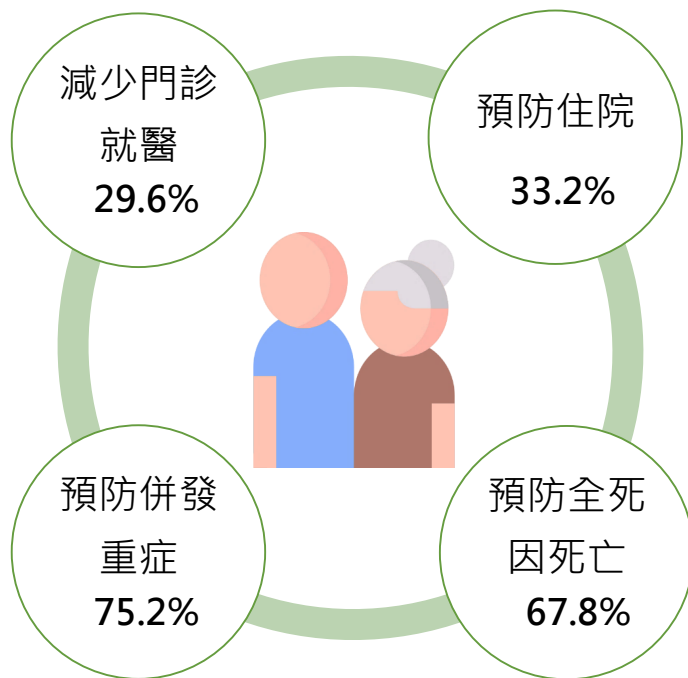
臺灣流感病毒流行狀況 2018-2019

疾病管制署流感速訊 (2019年第12週)



流感疫苗對長者具中高度保護力

國內研究顯示，接種流感疫苗顯著減少長者因實驗室確診流感之疾病負擔。

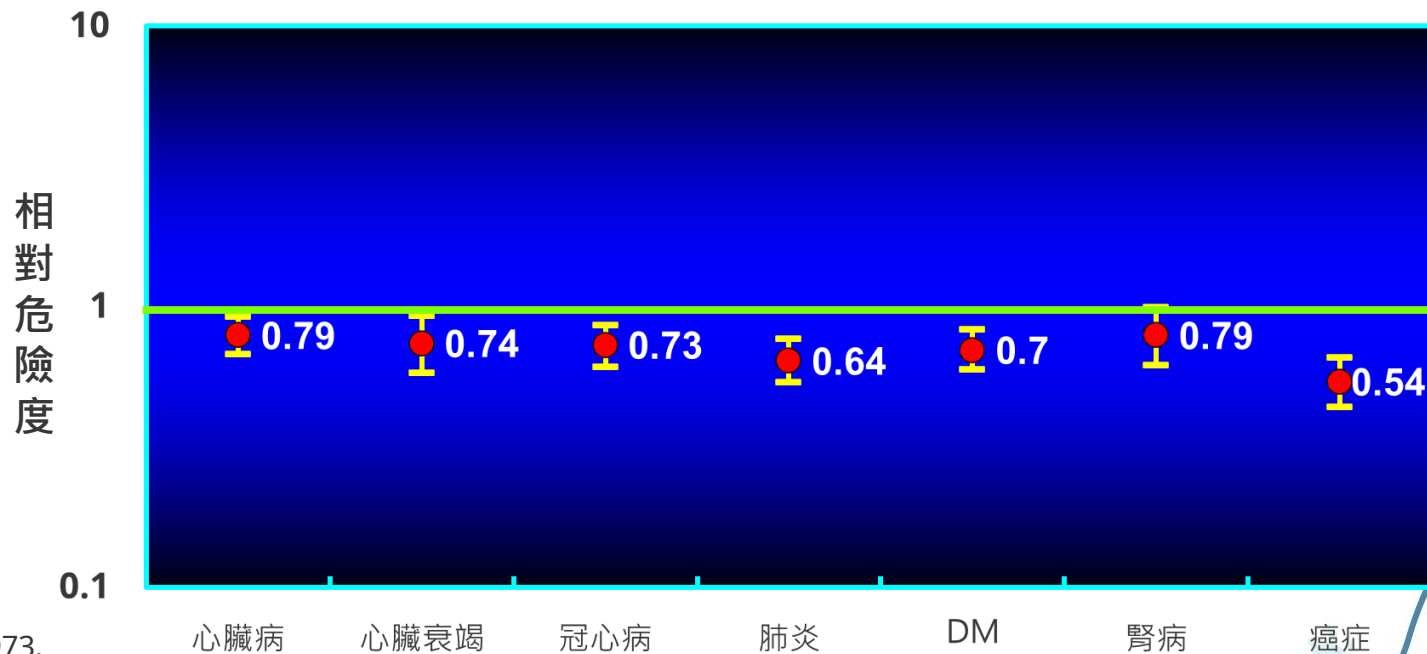


Vaccine 2025; 64: 127690

更新簡報P.27數據

長者接種流感疫苗降低死亡率

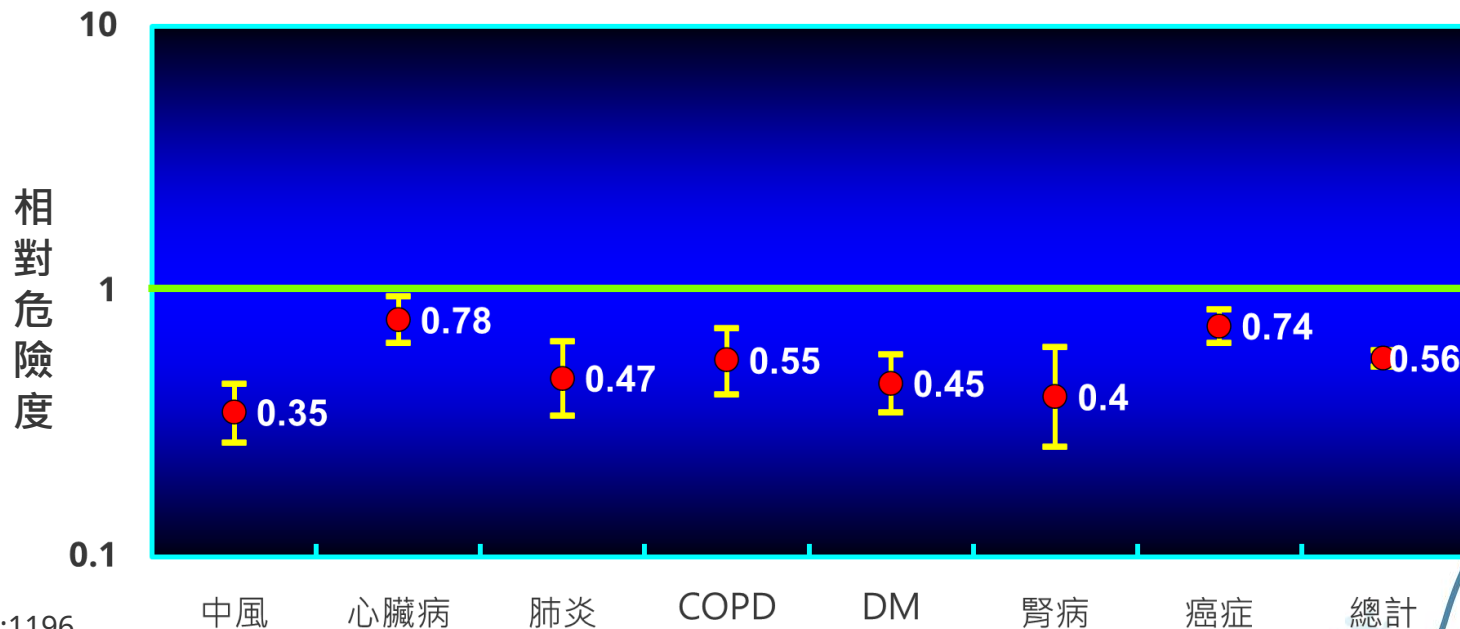
2000-2001 新北市， ≥ 65 歲老人，N=226,997



資料來源：
Wang ST. Vaccine 2005;23:1973.

長者接種流感疫苗降低死亡率

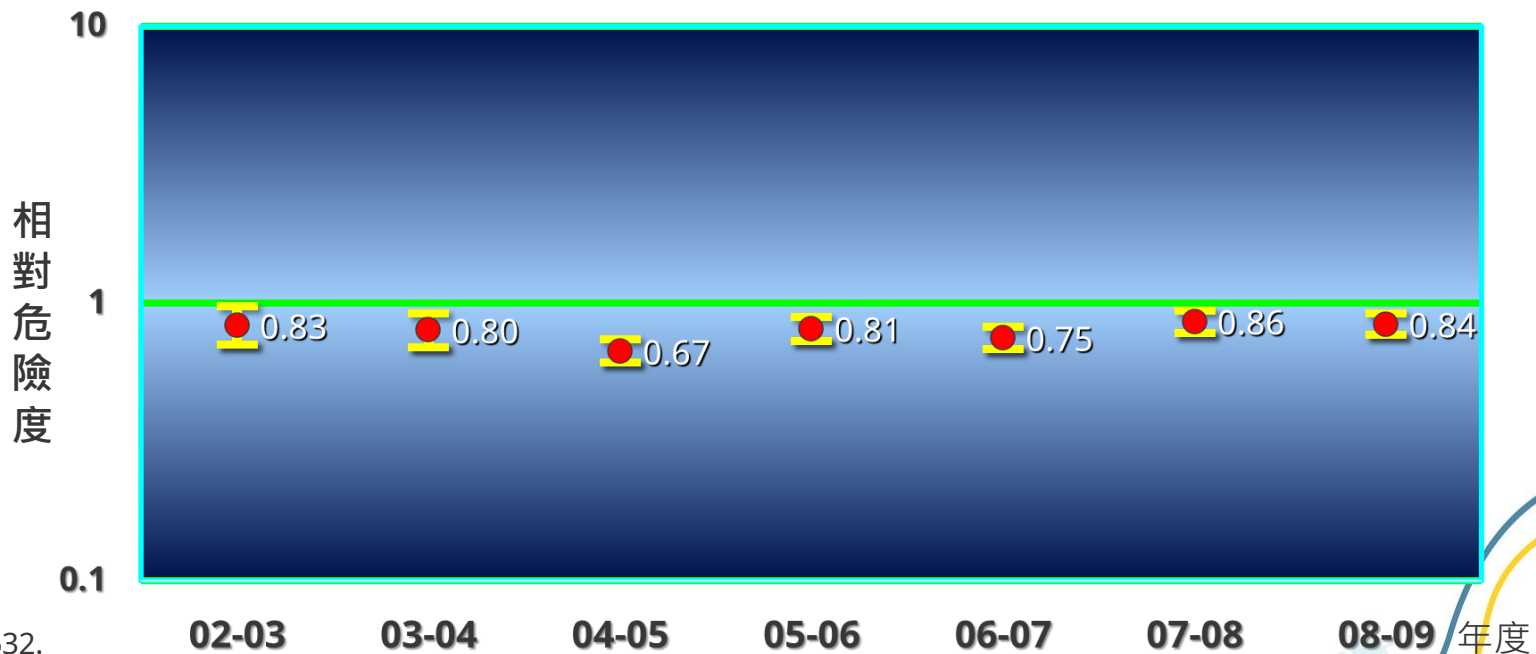
2000-2001 高雄縣， ≥ 65 歲老人， $N=35,637$



資料來源：
Wang CS. Vaccine 2007;25:1196.

臺灣長者接種流感疫苗的保護效益

N=1,000,000, Age≥65 yrs, Taiwan, 2002-2009

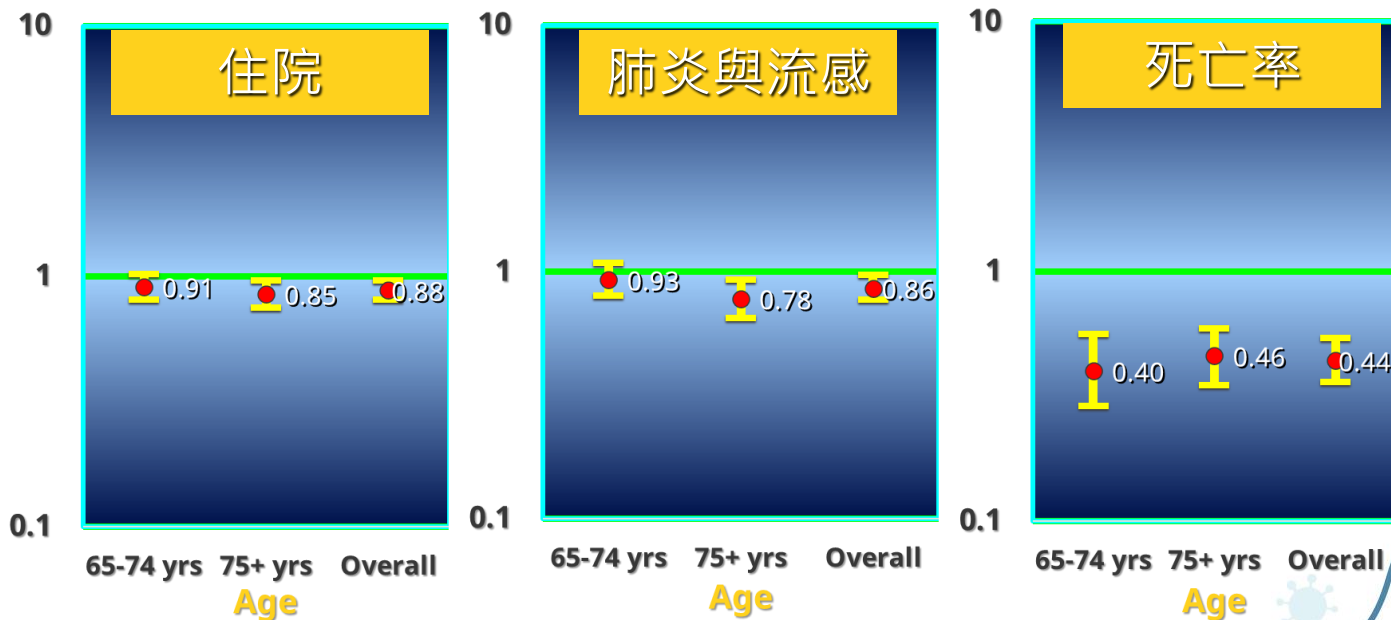


資料來源：
Chiu PJ. Vaccine 2013;31:632.

臺灣糖尿病長者接種流感疫苗的效益

N=9,025, Age≥65 yrs, Taiwan, 2002-2009

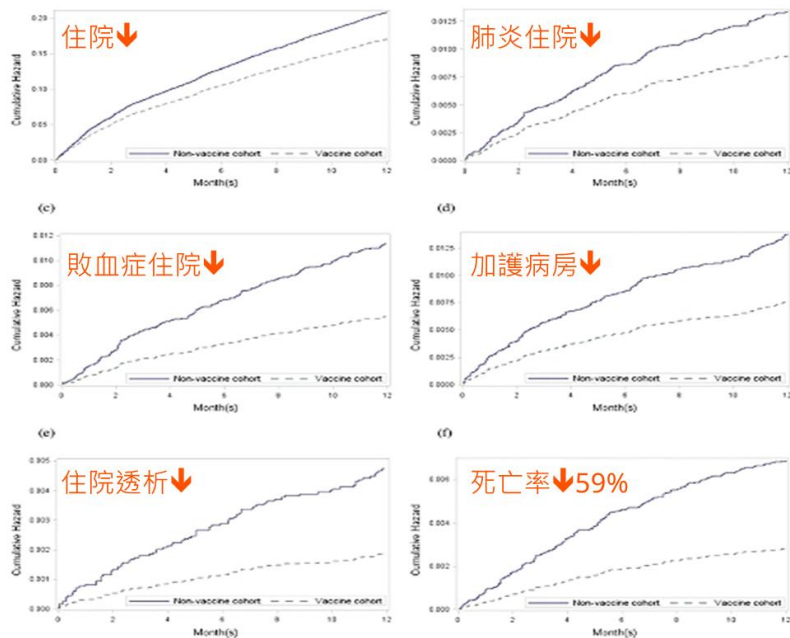
調整後相對危險性



資料來源：
Wang IK. Vaccine 2013;31:718.

Influenza vaccination in SLE patients

National Health Insurance Database, Taiwan, 2001-2012, N= 1765 (vaccine+) + 8360 (vaccine-)



SLE:
Systemic lupus
erythematosus
全身性紅疹性狼瘡

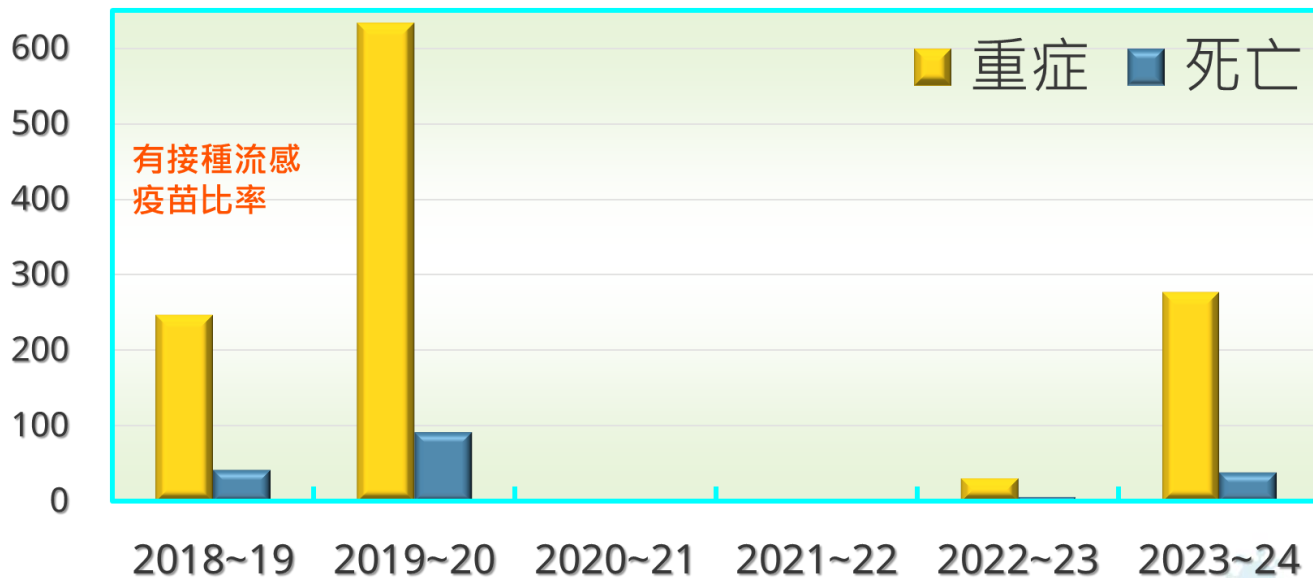
資料來源：Chang CC. Sci Rep 2016;6:37817.

流感併發重症病例數 與接種流感疫苗比率

臺灣疾病管制署，2018~2024.1.7



保護效益



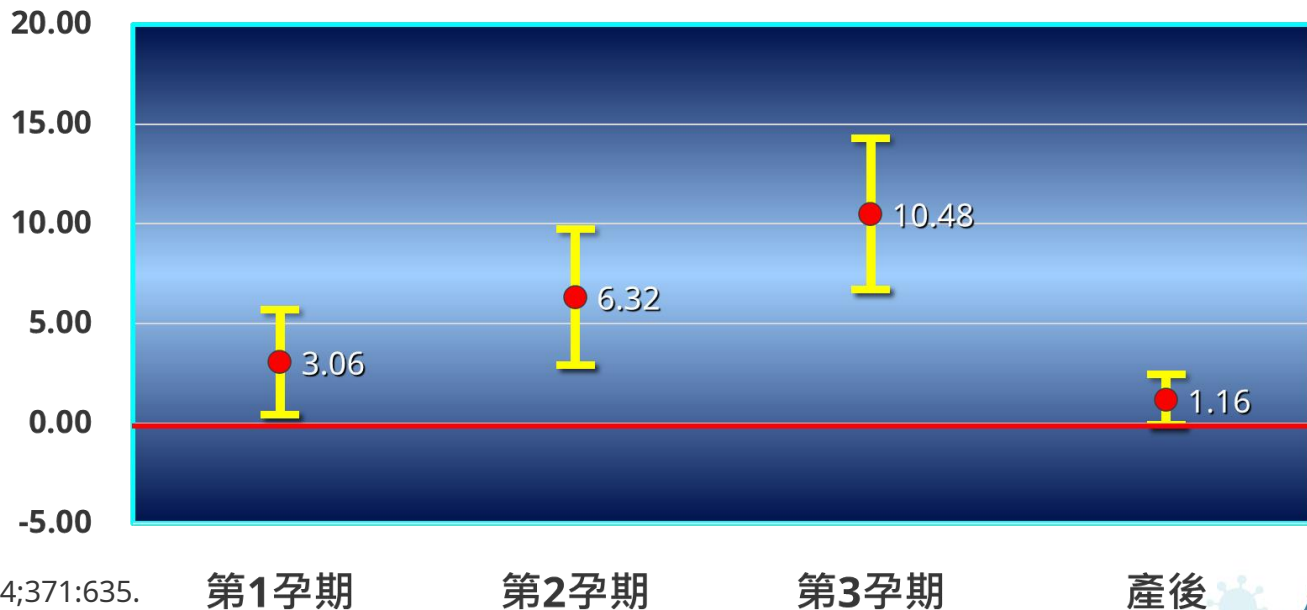
資料來源：
Darvishian M. Lancet Infect Dis
2014;14(12):1228.



孕婦得到流感的住院率

1974-1993, Tennessee Medicaid

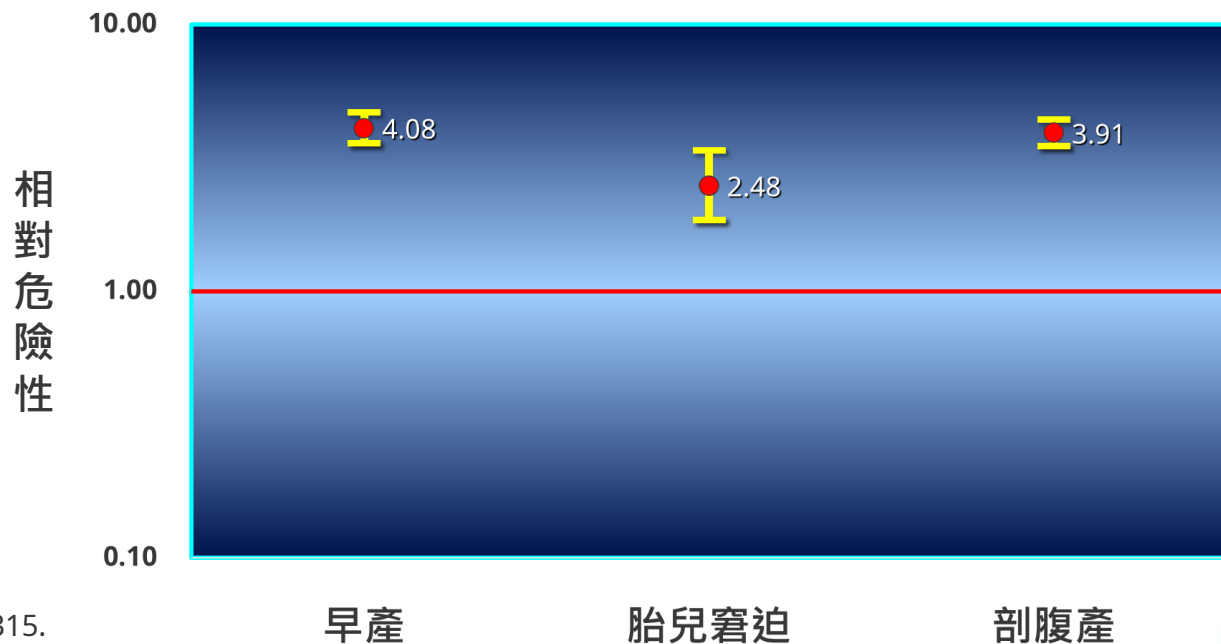
Excess admission (/10,000/month)



資料來源：
DiazGranados CA. N Engl J Med 2014;371:635.

孕婦得到流感對胎兒的影響

1998-2002, USA, N=6,277,508



資料來源：
Cox S. Obstet Gynecol 2006;107:1315.

無論孕期，孕婦接種流感疫苗不影響 早產與胎兒生長遲滯發生率 N=57554，2004-2009，美國CDC

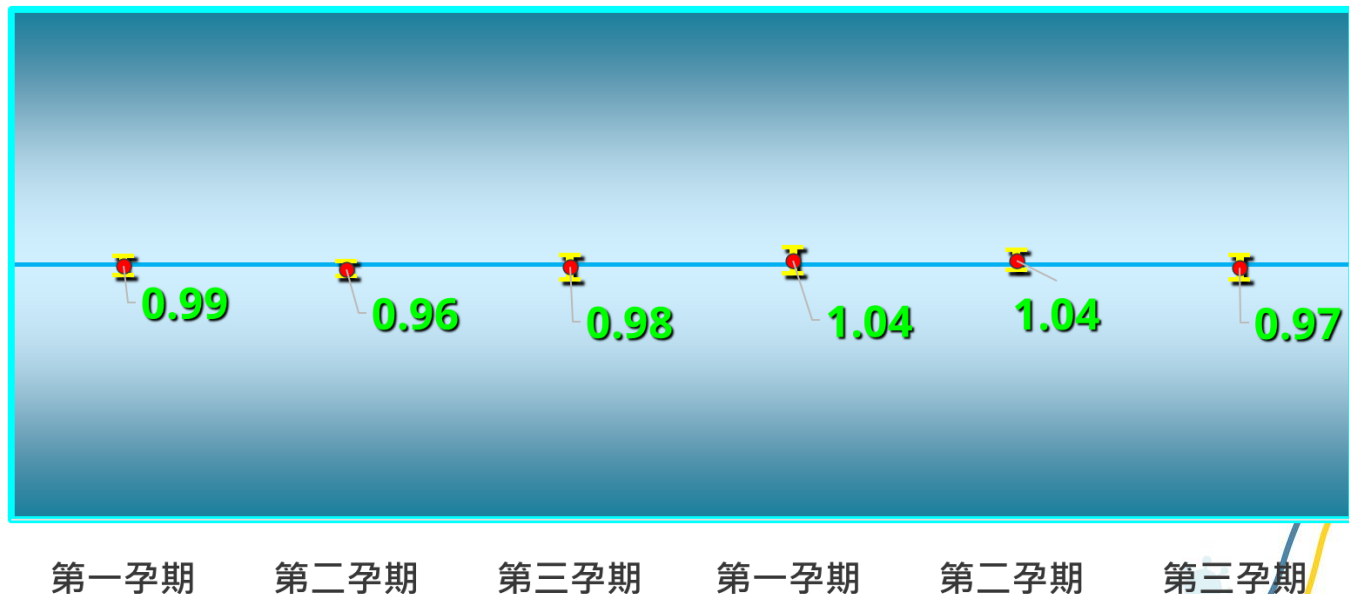


Odds ratio

10.0

1.0

0.1



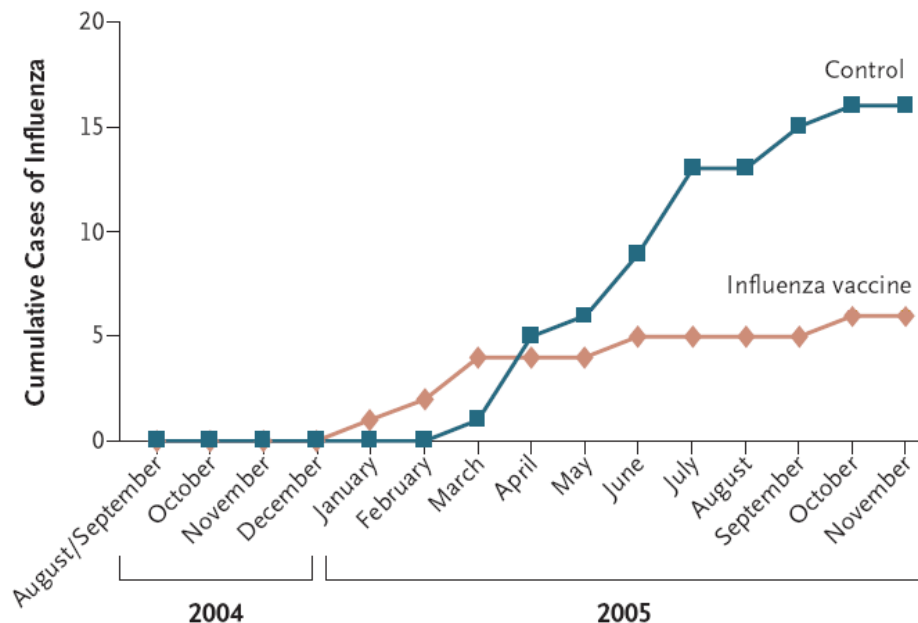
資料來源：
Nordin JD. Pediatr 2014;164:1051.



孕婦接種流感疫苗 (3rd trimester)

六個月以下嬰兒的保護效益

Bangladesh, N=340, 2004-2005



資料來源：
Zaman K. N Engl J Med 2008;359:1.



孕婦接種流感疫苗 (懷孕週數17-25 wks) 尼泊爾，2011-2013

■ 減少19%孕婦與36%
0-6個月嬰兒流感
發生率

■ 疫苗減少低體重新生兒
15%，對流產、死胎、
早產率沒有影響

	Cohort 1						Cohort 2						Combined					
	Placebo		Vaccine		Effect size (95% CI)†	p-value	Placebo		Vaccine		Effect size (95% CI)†	p-value	Placebo		Vaccine		Effect size (95% CI)†	p-value
	Cases*	Incidence†	Cases*	Incidence†			Cases*	Incidence†	Cases*	Incidence†			Cases*	Incidence†	Cases*	Incidence†		
Maternal																		
All laboratory-confirmed influenza	30 (639.4)	46.9	17 (662.4)	25.7	0.55 (0.39 to 0.99)	0.047	14 (454.0)	30.8	14 (449.4)	31.2	1.02 (0.48 to 2.18)	0.956	44 (1093.4)	40.3	31 (111.8)	27.9	0.69 (0.44 to 1.10)	0.122
Laboratory-confirmed influenza (pregnancy)	15 (260.6)	57.6	7 (271.3)	25.8	0.45 (0.18 to 1.10)	0.078	9 (159.8)	56.3	12 (160.7)	74.7	1.38 (0.56 to 3.37)	0.484	24 (420.5)	57.1	19 (435.9)	44.0	0.77 (0.42 to 1.43)	0.414
Laboratory-confirmed influenza (postpartum)	15 (378.9)	39.6	10 (391.1)	25.6	0.65 (0.28 to 1.47)	0.299	5 (294.1)	17.0	2 (288.7)	6.9	0.41 (0.08 to 2.09)	0.282	20 (672.9)	29.7	12 (679.9)	17.6	0.59 (0.28 to 1.23)	0.162
Infant																		
All influenza-like illness	409 (349.9)	1168.8	439 (363.6)	1207.4	1.04 (0.88 to 1.22)	0.674	328 (270.0)	1214.9	272 (258.7)	1047.5	0.87 (0.71 to 1.05)	0.145	737 (619.9)	1188.9	710 (622.3)	1140.9	0.96 (0.85 to 1.09)	0.556
Preterm (<37 weeks' gestation)	145 of (14%) 1023	NA	137 (13%) of 1027	NA	0.94 (0.76 to 1.17)	0.584	102 (13%) of 790	NA	88 (11%) of 782	NA	0.87 (0.67 to 1.14)	0.314	247 (14%) of 1813	NA	225 (12%) of 1809	NA	0.91 (0.77 to 1.08)	0.289
SGA† (intergrowth)	334 (45%) of 741	NA	327 (42%) of 764	NA	0.95 (0.85 to 1.06)	0.374	229 (41%) of 559	NA	211 of (39%) of 543	NA	0.95 (0.82 to 1.10)	0.475	563 (43%) of 1300	NA	538 (41%) of 1307	NA	0.95 (0.87 to 1.04)	0.121
SGA (Alexander)	400 (51%) of 775	NA	387 (48%) of 802	NA	0.93 (0.85 to 1.03)	0.182	275 (47%) of 580	NA	250 (44%) of 573	NA	0.92 (0.81 to 1.04)	0.198	675 (50%) of 1355	NA	637 (46%) of 1375	NA	0.93 (0.86 to 1.01)	0.068
Birthweight, g	2741.0 (465.3)	NA	2770.0 (440.6)	NA	28.9 (-15.6 to 73.5)	0.203	2789.8 (441.8)	NA	2851.8 (458.6)	NA	62.0 (-10.0 to 113.9)	0.029	2763.9 (455.9)	NA	2803.9 (449.9)	NA	42.1 (8.2 to 76.0)	0.035
Gestational age, weeks	39.4 (2.7)	NA	39.4 (2.6)	NA	0.002 (-0.23 to 0.23)	0.986	39.3 (2.4)	NA	39.5 (2.4)	NA	0.16 (-0.07 to 0.40)	0.168	39.4 (2.6)	NA	39.4 (2.5)	NA	0.07 (-0.09 to 0.24)	0.386

資料來源：
Stein Hoff MC. Lancet Infect Dis 2017;17: 981.

孕婦接種H1N1大流行流感疫苗

2009-2010 臺灣疫苗傷害救濟申訴與健保資料

- 接種疫苗：14,474名孕婦，自然流產發生率2.3% (95% CI, 1.4–3.8%)
- 未接種疫苗：自然流產發生率12.8% (95% CI, 12.8–12.9%)

Table 2. Adverse events following 2009 H1N1 vaccines among pregnant women, November 1, 2009–August 31, 2010, by vaccine type and timing of vaccination.

Adverse event	Timing of vaccination ^a					
	First trimester		Second trimester		Third trimester	
	Unadjuvanted vaccine (n = 10)	MF59®-adjuvanted vaccine (n = 4)	Unadjuvanted vaccine (n = 15)	MF59®-adjuvanted vaccine (n = 1)	Unadjuvanted vaccine (n = 5)	MF59®-adjuvanted vaccine (n = 2)
Pregnancy-specific						
Spontaneous abortion	9	1	6	0	0	0
Stillbirth	0	0	7	1	2	1
Neonatal death	0	0	2 ^b	0	1 ^c	1 ^d
Nonpregnancy-specific	1 ^e	1 ^f	0	0	2 ^f	0
Inadvertent immunization	0	2 ^g	0	0	0	0

資料來源：Huang WT. PLoS ONE 2011;6(8): e23049.

一位一個月大男嬰

2019.3.5



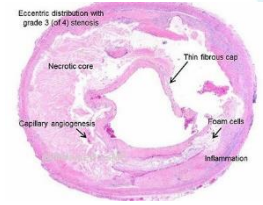
- 2019.3.4：輕微咳嗽
- 2019.3.5：意識不清，活力減弱，送至急診
 - 胸部瀰漫性細囉音，capillary refilling time延長
 - VBG: pH 7.133; pCO₂ 74.7; pO₂ 57.8; HCO₃ 24.5
 - 流感快篩A型流感陽性
- TOCC: 母親與哥哥流感快篩A型流感陽性
- 全家未接種疫苗：竹科工程師，「崇尚自然」
- 治療：peramivir，vancomycin，cefotaxime



Photo's taken by Dr. Ping-Ing Lee



Atherosclerosis with inflammation

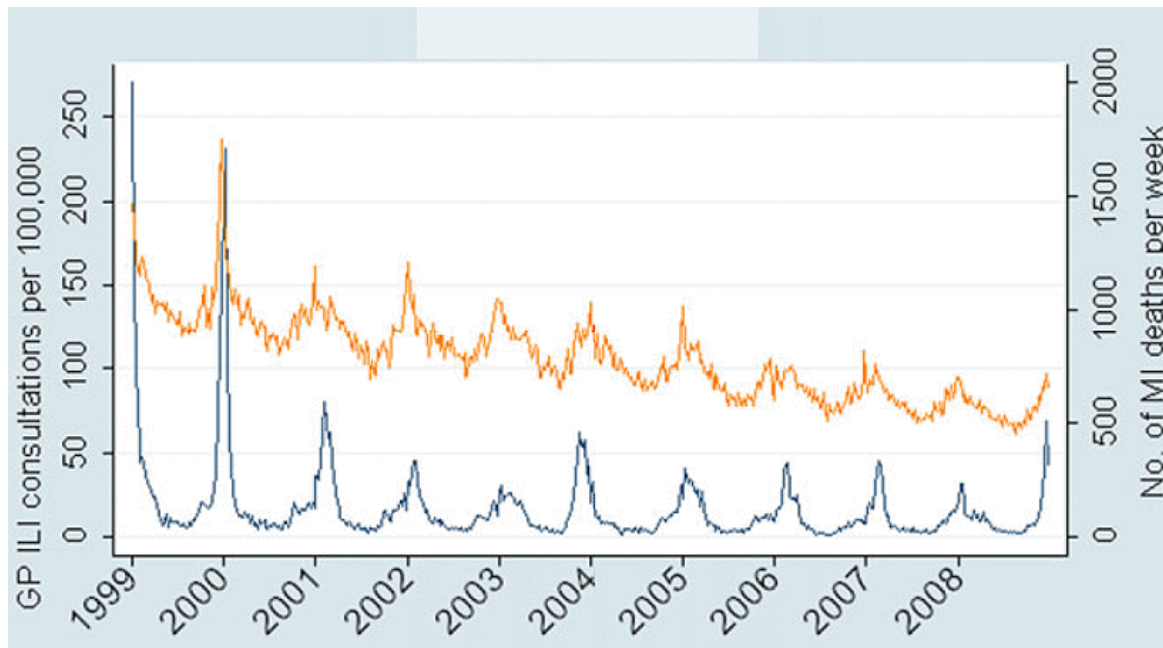


- A **consequence** or part of the **pathogenesis** of the disease?
- Rayer, **1823**: Morbid ossification of the arteries (calcification of atherosclerosis) [is] **the result of the inflammation** of their fibrous layer.
- Virchow, 1859: an inflammation of the inner arterial coat to be the **starting point** of the so-called atheromatous degeneration.

Source of image: Mayo Clinic on Twitter
資料來源：Nieto FJ. Am J Epidemiol 1998;148(10):937.

Seasonality of influenza-like-illness myocardial infarction

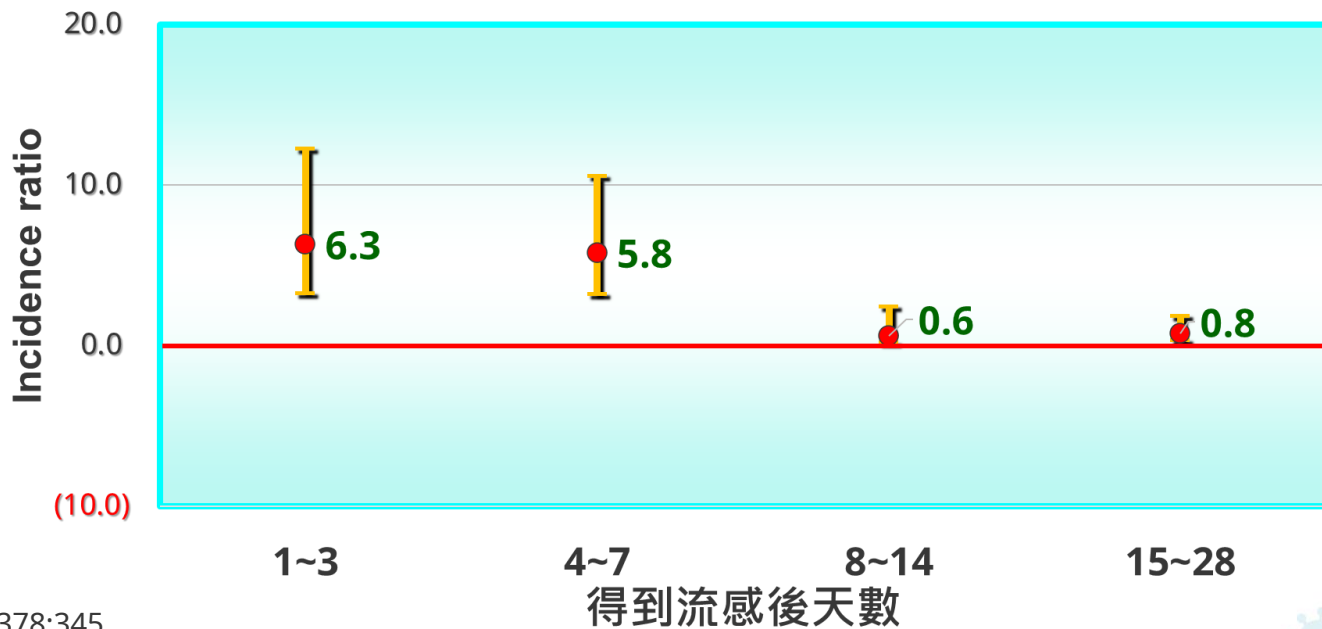
1998~2008, England and Wales



資料來源：
Warren-Gash C. J Infect Dis
2011;203(12):1710.

流感7天內 急性心肌梗塞發生率增加6倍

≥35歲，N=364，加拿大，2018

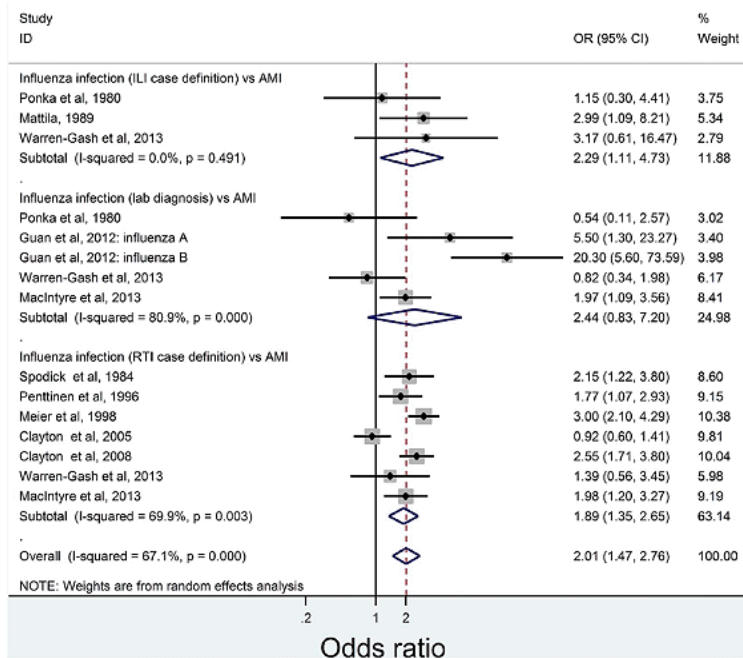


資料來源：
Kwong JC. N Engl J Med 2018;378:345.

流感增加急性心肌梗塞發生率 綜合分析，N=10，2015



Pooled OR: **2.01**
(95% CI 1.47 to 2.76)



資料來源：
Barnes M. Heart 2015;101(21):1738.

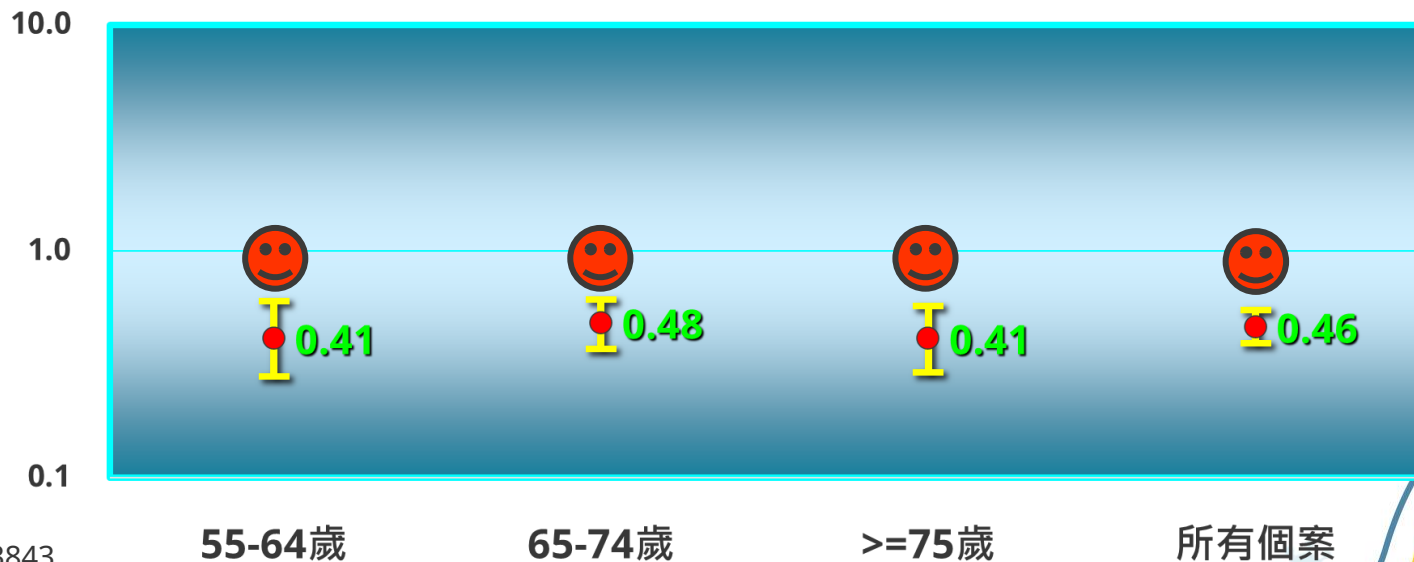


流感疫苗減少慢性阻塞性肺病長者 發生冠狀動脈疾病

N = 7722 , ≥55歲 , 臺灣 , 2000-2007

Hazard ratio

😊 P < 0.05

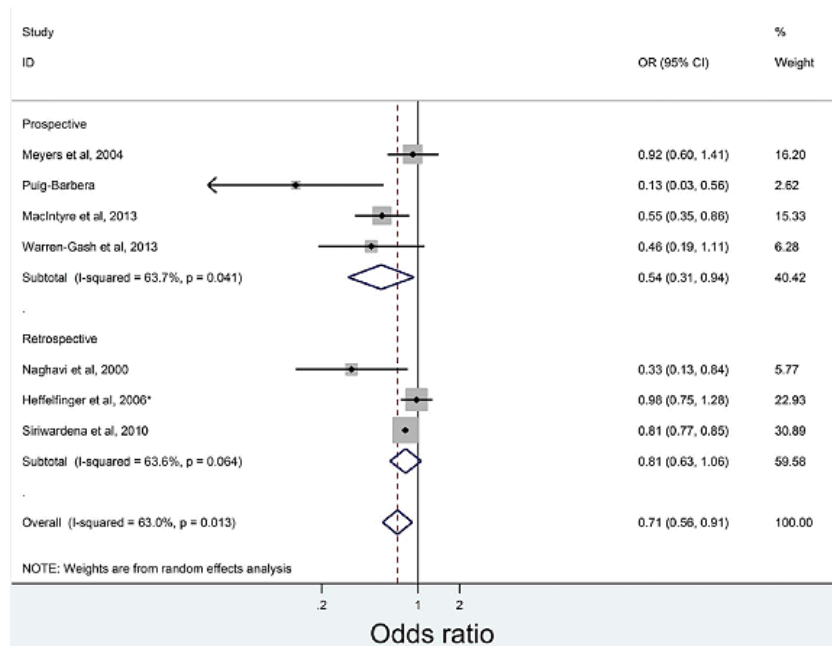


資料來源：
Sung LC. Vaccine 2014;32(30):3843.

流感疫苗減少急性心肌梗塞發生率 綜合分析，N=8，2015



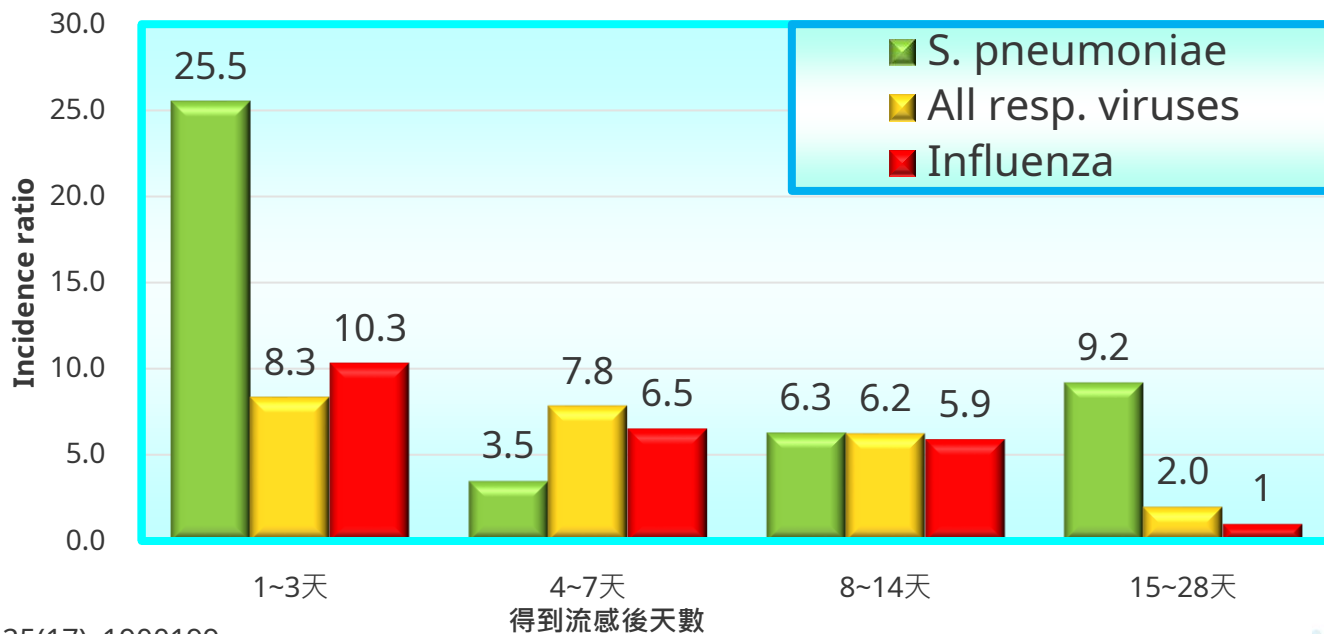
Pooled OR: **0.71**
(95% CI 0.56~0.91)



資料來源：
Barnes M. Heart 2015;101(21):1738.

各種感染症與腦中風發生率相關性

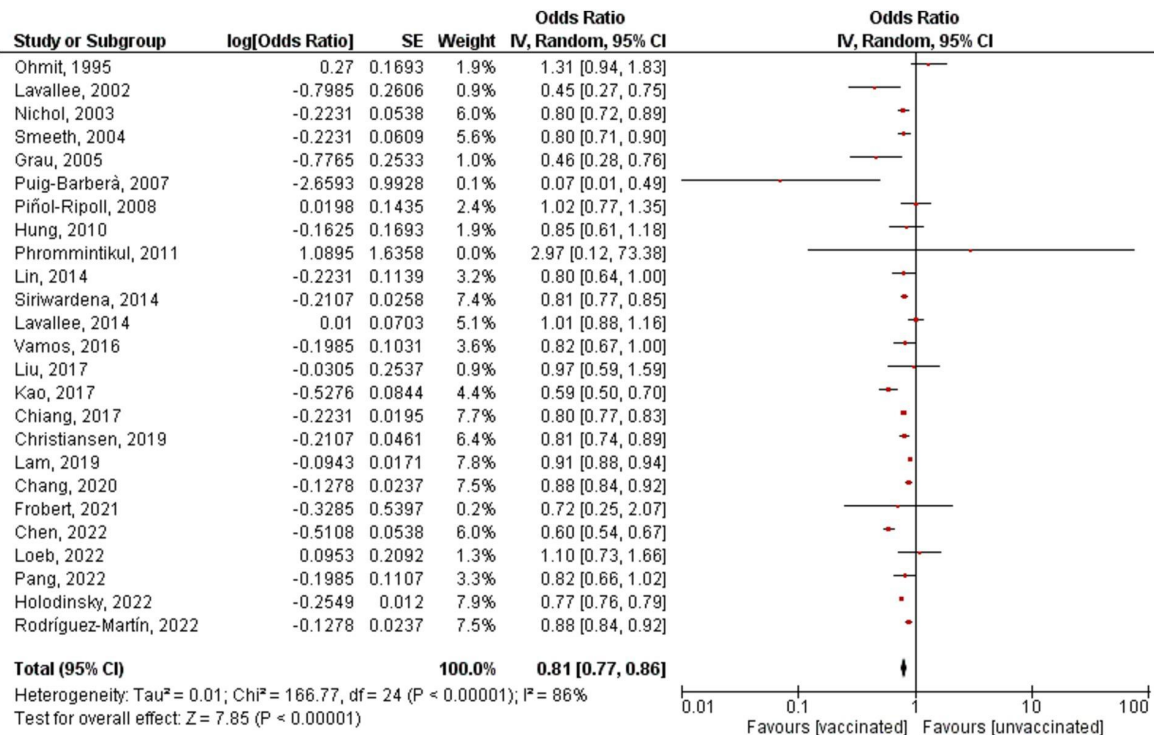
≥40歲，丹麥，2000~2009



資料來源：
Ohland J. Euro Surveill 2020; 25(17): 1900199.

流感疫苗降低中風發生率

多國研究接種流感疫苗者比未接種者中風（包括住院或發生）機率降低約19%（OR = 0.81 95% CI: 0.77–0.86）



流感疫苗降低心血管疾病與腦血管 疾病發生率

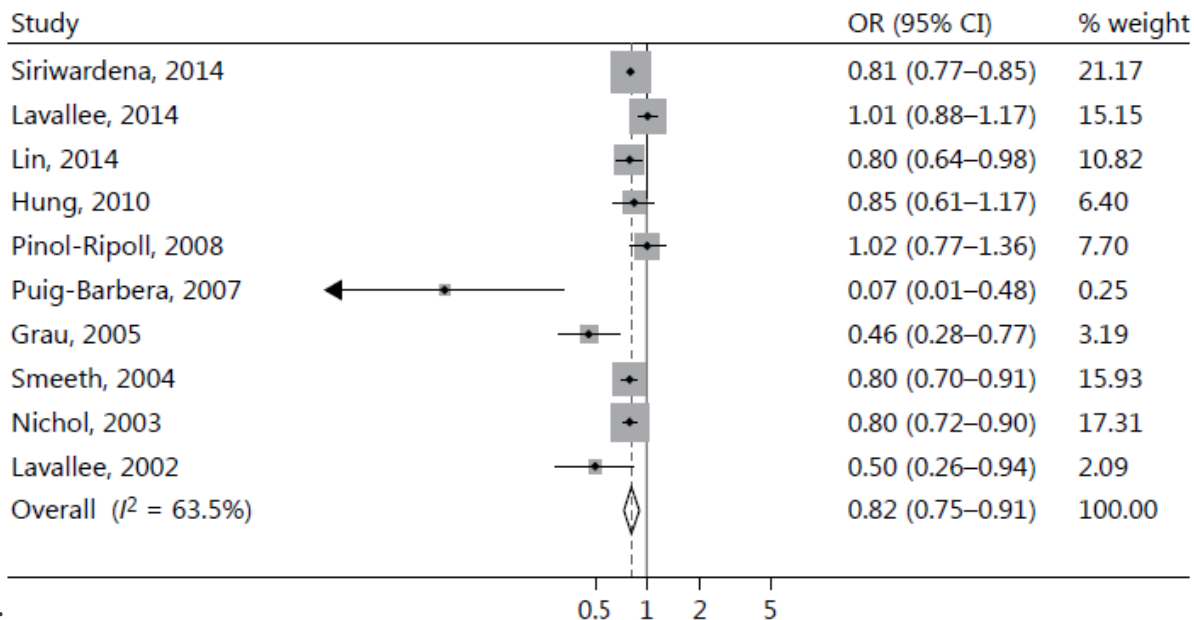
N = 286,383, ≥65歲, 1998~2000, 美國



資料來源：
Nichol KL. N Engl J Med 2003;348:1322.

流感疫苗減少腦中風發生率 綜合分析，N=11，2017

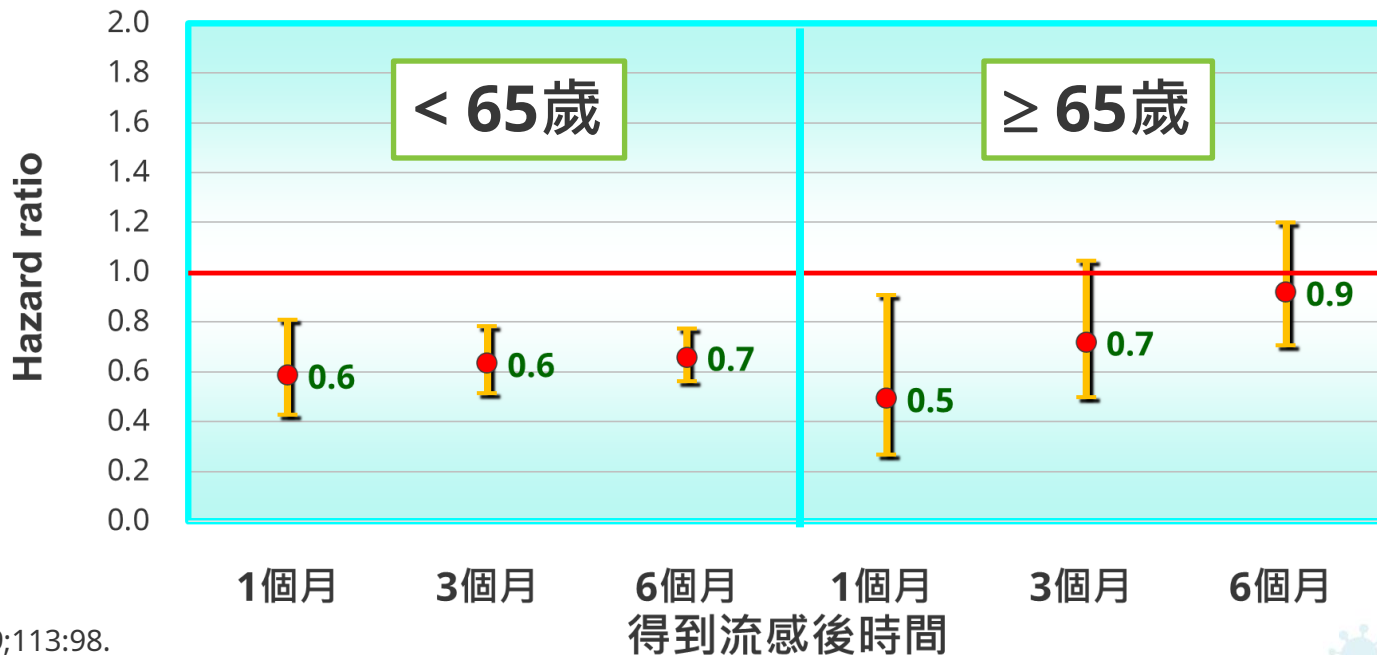
Pooled odds
ratio: **0.82** (95%
CI 0.75 to 0.91)



資料來源：
Lee KR. Neuroepidemiology 2017;48:103.

以oseltamivir治療流感 可降低腦中風發生率

≥18歲，N=151,930，2000~2006，美國



資料來源：
Madjid M. Cardiology 2009;113:98.

FluMist Quadrivalent

MedImmune, Inc., Maryland



- Intranasally administered live-attenuated influenza vaccine: FDA approval June 17, 2003.6.
- Adverse reactions: running nose (20-78%), headache (2-46%), sorethroat (15-27%), fever (0-26%), vomiting (3-13%)
- Advantage: local immune response, high acceptability



FluMist Quadrivalent

MedImmune, Inc., Maryland



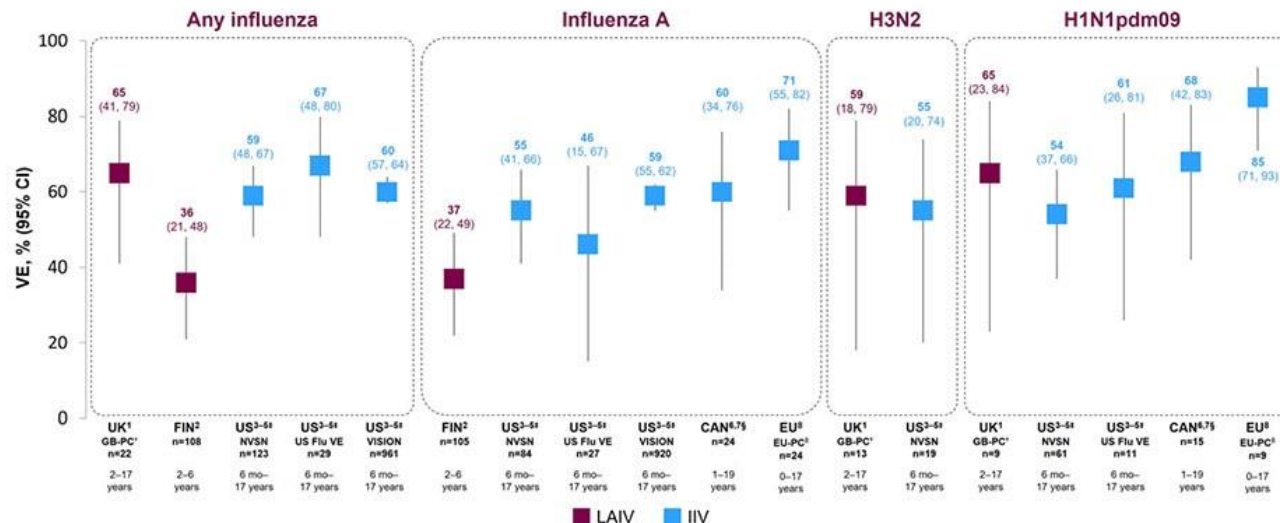
■ Disadvantage

- Indicated only for healthy, nonpregnant persons aged 2 through 49 years
- Not recommended for persons who care for severely immunosuppressed persons
- Expensive



活性減毒與非活性流感疫苗的保護效益差不多

Figure 1. Influenza VE in outpatient/primary care in children during the 2023/24 influenza season



¹96% of controls received LAIV. ²Data for all vaccines, but little use of quadrivalent LAIV or trivalent IIV in the US. ³⁻⁵Of publicly funded vaccines, > 98% were egg-based IIV. ⁶Vaccine brand information available for 77% of vaccinated controls, all vaccines were quadrivalent: 9% cell-based, 1% LAIV, 5% adjuvanted vaccine, 12% high-dose vaccine, 73% unadjuvanted, standard dose, egg-propagated, inactivated. CAN, Canada; CI, confidence interval; EN-H, England hospital; EU, Europe; EU-PC, Europe primary care; FIN, Finland; GB-PC, Great Britain primary care; IIV, inactivated influenza vaccine; LAIV, live attenuated influenza vaccine; mo, months; n, number of vaccinated cases; NVSN, New Vaccine Surveillance Network; VE, vaccine effectiveness.

1. Whitaker H et al. *medRxiv*. 2024.02.27.24303426; 2. Nohynek H, Ikonen N, Baum U. Poster presented at: ESCMID Global 2024; April 27–30, 2024; 3. Frutos AM et al. *MMWR Morb Mortal Wkly Rep*. 2024 Feb 29;73(8):168–174; 4. Grohskopf LA et al. *MMWR Recomm Rep*. 2023;72(2):1–25; 5. Centers for Disease Control and Prevention. Live Attenuated Influenza Vaccine [LAIV] (The Nasal Spray Flu Vaccine); 6. Skowronski DM et al. *Euro Surveill*. 2024;29(7):2400076; 7. Government of Canada. Statement on seasonal influenza vaccine for 2023–2024. Accessed April 8, 2024; 8. Maurel M et al. *Euro Surveill*. 2024;29(8):2400069.

Contraindications/precautions of influenza vaccines

ACIP, USA, 2022

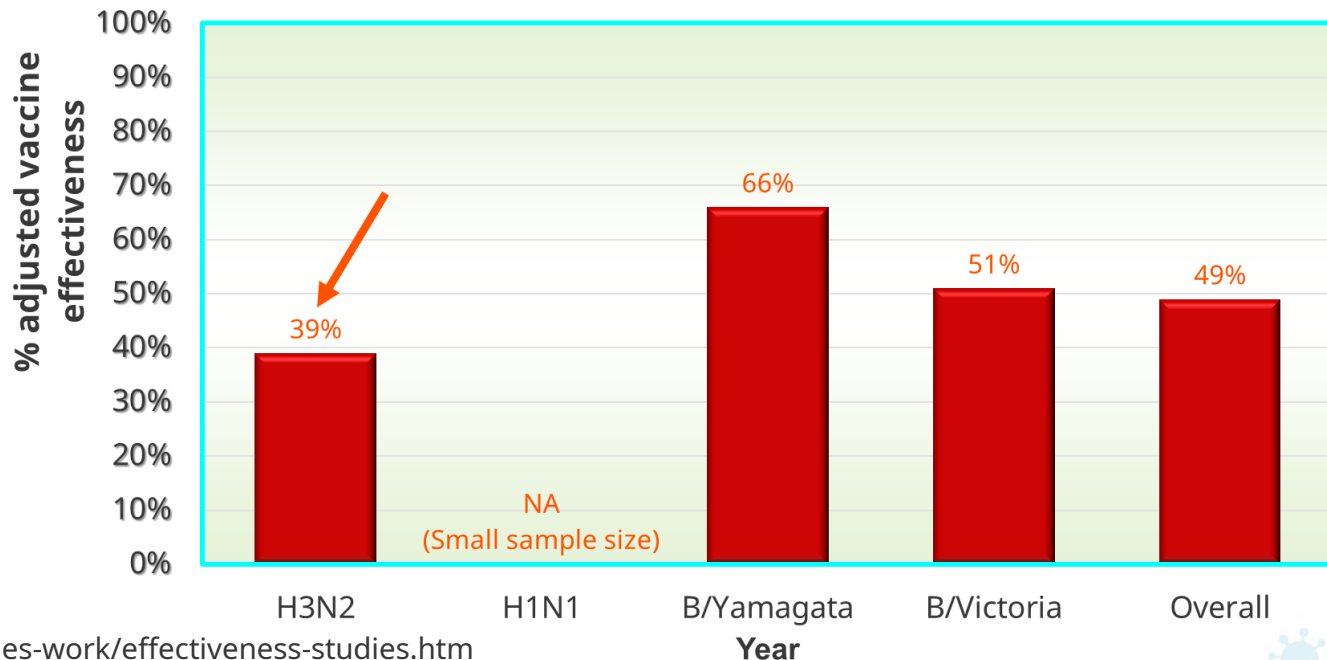
TABLE 2. Contraindications and precautions for the use of influenza vaccines — United States, 2022–23 influenza season*

Vaccine type	Contraindications	Precautions
Egg-based IIV4s	History of severe allergic reaction (e.g., anaphylaxis) to any component of the vaccine[†] or to a previous dose of any influenza vaccine (i.e., any egg-based IIV, cclIV, RIV, or LAIV)[§]	- Moderate or severe acute illness with or without fever - History of Guillain-Barré syndrome within 6 weeks of receipt of influenza vaccine
cclIV4	History of severe allergic reaction (e.g., anaphylaxis) to a previous dose of any cclIV or any component of cclIV4[§]	- Moderate or severe acute illness with or without fever - History of Guillain-Barré syndrome within 6 weeks of receipt of influenza vaccine - History of severe allergic reaction to a previous dose of any other influenza vaccine (i.e., any egg-based IIV, RIV, or LAIV)[†]
RIV4	History of severe allergic reaction (e.g., anaphylaxis) to a previous dose of any RIV or any component of RIV4[§]	- Moderate or severe acute illness with or without fever - History of Guillain-Barré syndrome within 6 weeks of receipt of influenza vaccine - History of severe allergic reaction to a previous dose of any other influenza vaccine (i.e., any egg-based IIV, cclIV, or LAIV)[†]
LAIV4	- History of severe allergic reaction (e.g., anaphylaxis) to any component of the vaccine[†] or to a previous dose of any influenza vaccine (i.e., any egg-based IIV, cclIV, RIV, or LAIV)[§] - Concomitant aspirin- or salicylate-containing therapy in children and adolescents[§] - Children aged 2 through 4 years who have received a diagnosis of asthma or whose parents or caregivers report that a health care provider has told them during the preceding 12 months that their child had wheezing or asthma or whose medical record indicates a wheezing episode has occurred during the preceding 12 months - Children and adults who are immunocompromised due to any cause, including but not limited to immunosuppression caused by medications, congenital or acquired immunodeficiency states, HIV infection, anatomic asplenia, or functional asplenia (e.g., due to sickle cell anemia) - Close contacts and caregivers of severely immunosuppressed persons who require a protected environment - Pregnancy - Persons with active communication between the CSF and the oropharynx, nasopharynx, nose, or ear or any other cranial CSF leak - Persons with cochlear implants^{**} - Receipt of influenza antiviral medication within the previous 48 hours for oseltamivir and zanamivir, previous 5 days for peramivir, and previous 17 days for baloxavir^{††}	- Moderate or severe acute illness with or without fever - History of Guillain-Barré syndrome within 6 weeks of receipt of influenza vaccine - Asthma in persons aged ≥5 years - Other underlying medical conditions that might predispose to complications after wild-type influenza infection (e.g., chronic pulmonary, cardiovascular [except isolated hypertension], renal, hepatic, neurologic, hematologic, or metabolic disorders [including diabetes mellitus])

資料來源：
Grohskopf LA. MMWR 2022;71(1):1.

CDC seasonal flu vaccine effectiveness studies

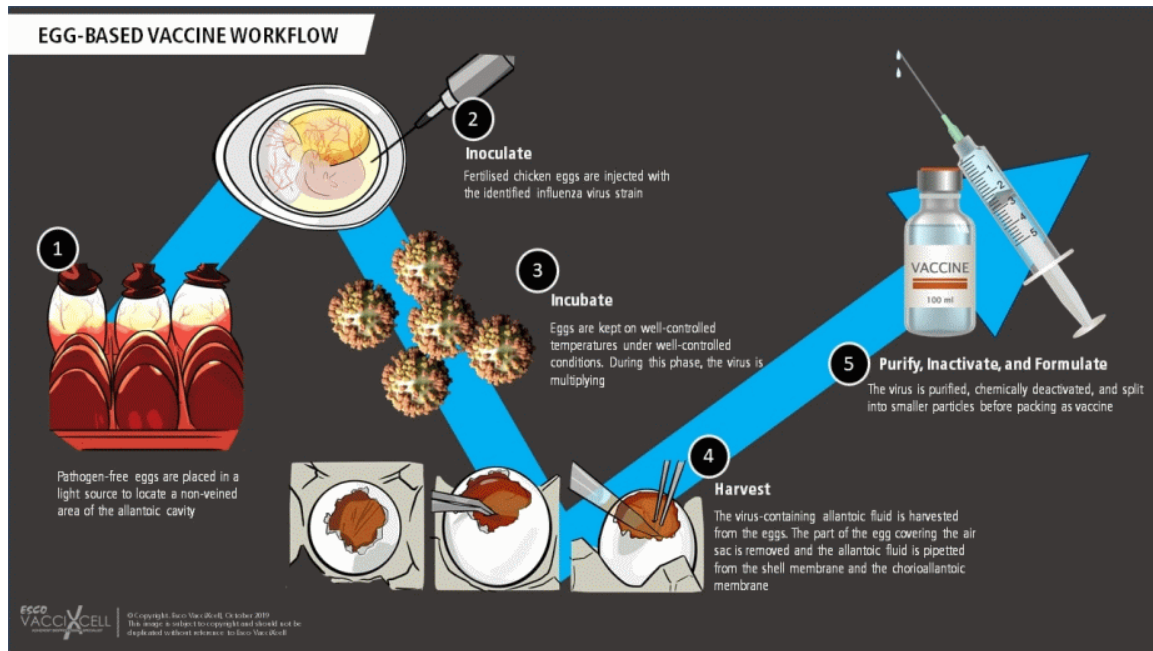
2012-2013, USA



資料來源：
<https://www.cdc.gov/flu/vaccines-work/effectiveness-studies.htm>

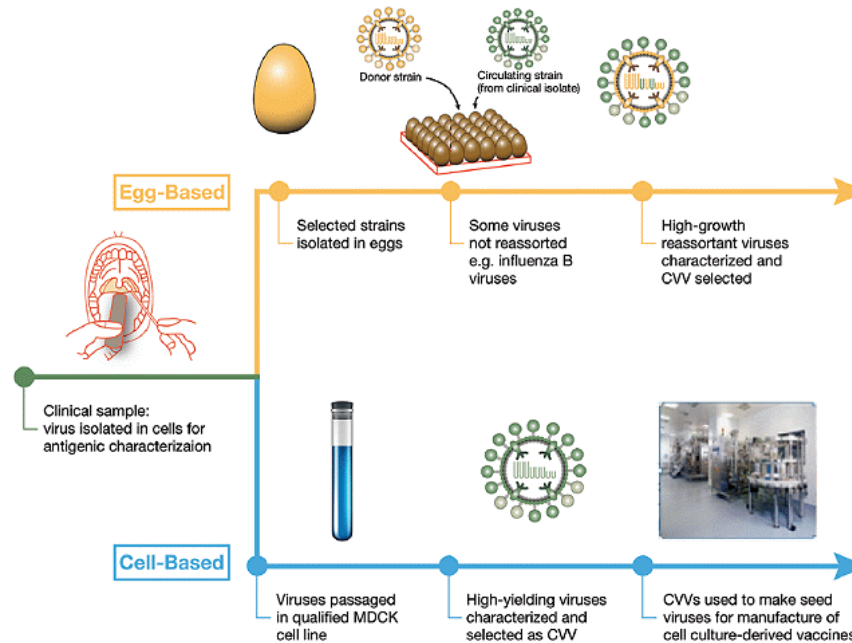
雞胚培養流感疫苗

Avian
influenza-like
virus?



資料來源：<https://escovaccixcell.com/news/is-your-flu-shot-an-egg-beater-or-a-cell-seeker>

Traditional egg-based and emerging cell-based manufacturing processes for influenza vaccines



資料來源：
Rajaram S. Ther Adv Vaccines Immunother
2020;8:2515135520908121.

Comparison between egg-based and cell-based influenza vaccines

July 28, 2019, Pharmacy Times

	Advantages	Disadvantages
Egg-based vaccine	• Low cost • Infrastructure	• Reliance on egg availability • Viral mutation • Slower manufacturing process
Cell-based vaccine	• No egg utilization • Quicker production • Less viral mutation	• Lack of developed infrastructure

資料來源：<https://www.pharmacytimes.com/publications/supplements/2019/July2019/cracking-the-code-of-optimized-influenza-prevention-cellbased-influenza-vaccine-advancement>

Recommended composition of influenza virus vaccines for use in the 2020 - 2021 northern hemisphere influenza season

Feb. 28, 2020, WHO



It is recommended that **quadrivalent vaccines** for use in the 2020 - 2021 northern hemisphere influenza season contain the following:

Egg-based Vaccines

- an A/Guangdong-Maonan/SWL1536/2019 (H1N1)pdm09-like virus;
- an A/Hong Kong/2671/2019 (H3N2)-like virus;
- a B/Washington/02/2019 (B/Victoria lineage)-like virus; and
- a B/Phuket/3073/2013 (B/Yamagata lineage)-like virus.

Cell- or recombinant-based Vaccines

- an A/Hawaii/70/2019 (H1N1)pdm09-like virus;
 - an A/Hong Kong/45/2019 (H3N2)-like virus;
 - a B/Washington/02/2019 (B/Victoria lineage)-like virus; and
 - a B/Phuket/3073/2013 (B/Yamagata lineage)-like virus.
- 

資料來源：
<https://www.who.int/influenza/vaccines/virus/recommendations/en/>



Adverse events of QIVc: mild to moderate in adults

N=2680, adults ≥ 18 years, 2016



*c: cell-based



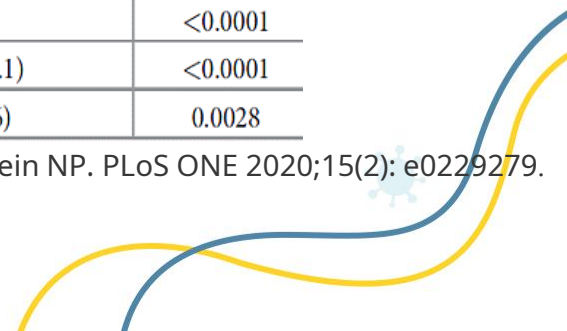
資料來源：
Bart S. Hum Vaccin Immunother
2016;12(9):2278.

Vaccine effectiveness of cell culture and egg-based inactivated influenza vaccines among 4-64-year-olds 2017-2018, California



Comparison	Vaccine Effectiveness (VE) Against Influenza A		
	Relative VE (%) (95% CI)	Absolute VE (%) (95% CI)	P-value
ccIIV4 vs SD-IIV3/4 vaccinees	8.0 (-10, 23)	n/a	0.35
ccIIV4 vaccinees vs unvaccinated	n/a	31.7 (18.7, 42.6)	<0.0001
SD-IIV3/4 vs unvaccinated	n/a	20.1 (14.5, 25.4)	<0.0001
Comparison	Vaccine Effectiveness (VE) Against Influenza B		
	Relative VE (%) (95% CI)	Absolute VE (%) (95% CI)	P-value
ccIIV4 vs SD-IIV3 vaccinees	39.6 (27.9, 49.3)	n/a	<0.0001
ccIIV4 vaccinees vs unvaccinated	n/a	40.9 (30.0, 50.1)	<0.0001
SD-IIV3 vaccinees vs unvaccinated	n/a	9.7 (3.5, 15.6)	0.0028

資料來源：Klein NP. PLoS ONE 2020;15(2): e0229279.



Protective efficacy of cell- and egg-based trivalent inactivated influenza virus vaccines in adults

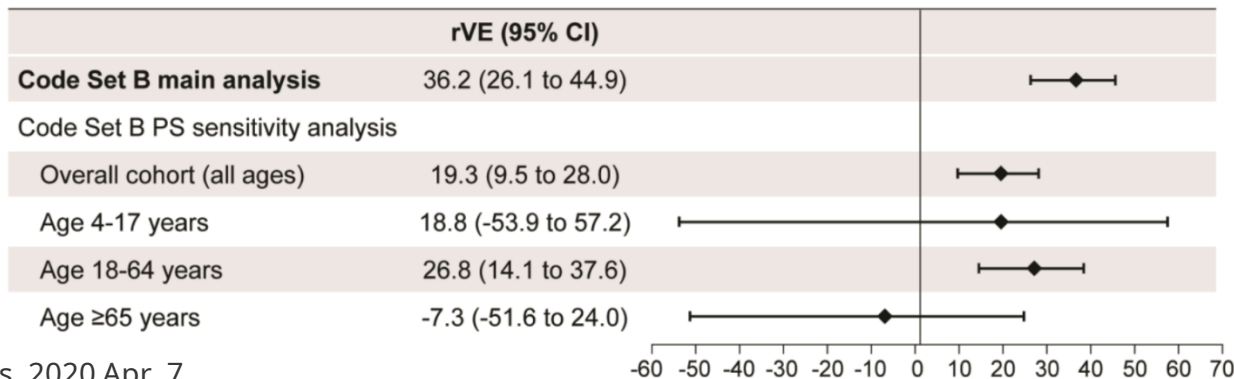
Strains	TIVc (n = 3776 ^a)		TIVe (n = 3638 ^a)		Placebo (n = 3843 ^a)		Vaccine efficacy ^b			
	No. of cases	Attack rate (%)	No. of cases	Attack rate (%)	No. of cases	Attack rate (%)	TIVc vs. placebo		TIVe vs. placebo	
							%	Lower limit of 97.5% CI ^c	%	Lower limit of 97.5% CI ^c
Antigenically matched^d (i.e. vaccine-like) strains										
Overall ^e	7	0.19	9	0.25	44	1.14	83.8**	61.0	78.4*	52.1
A/H1N1	5	0.13	8	0.22	43	1.12	88.2**	67.4	80.3*	54.7
A/H3N2 ^f	2	0.05	1	0.03	0	0				
B ^f	0	0	0	0	1	0.03				
Non-antigenically matched^g (i.e. non-vaccine-like) strains										
Overall	30	0.79	29	0.80	74	1.93	58.7	33.5	58.6	32.9
A/H1N1	1	0.03	0	0	8	0.21	87.3	4.6	100	33.9
A/H3N2	0	0	2	0.05	8	0.21	100	36.3	73.6	– 30.1
B	29	0.77	27	0.74	59	1.54	50.0	17.5	51.7	19.4
All culture-confirmed influenza										
Overall	42	1.11	49	1.35	140	3.64	69.5**	55.0	63.0*	46.7
A/H1N1	6	0.16	10	0.27	57	1.48	89.3**	73.0	81.5**	60.9
A/H3N2	6	0.16	12	0.33	25	0.65	75.6	35.1	49.3	– 9.0
B	30	0.79	27	0.74	61	1.59	49.9	18.2	53.2	22.2

Cell-based Egg-based

資料來源：Lamb YN. Drugs 2019;79(12):1337.

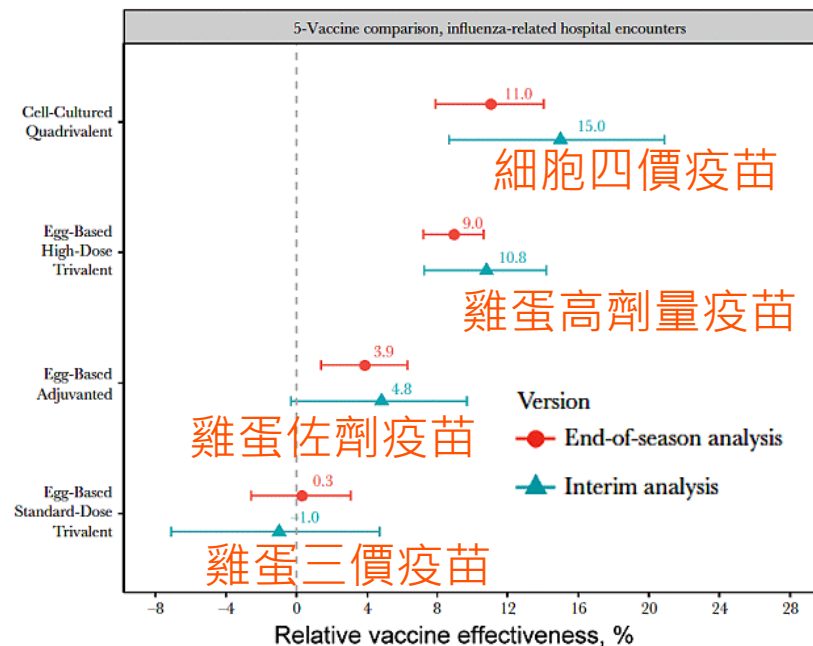
Relative effectiveness of cell-cultured and egg-based influenza vaccines among persons ≥ 4 years United States, 2017–2018

- Relative vaccine effectiveness: 36.2%, 95% CI (26.1, 44.9; $P < 0.001$)
- Total rVE was driven by the age group 18–64 years



資料來源：
Boikos C. Clin Infect Dis. 2020 Apr. 7

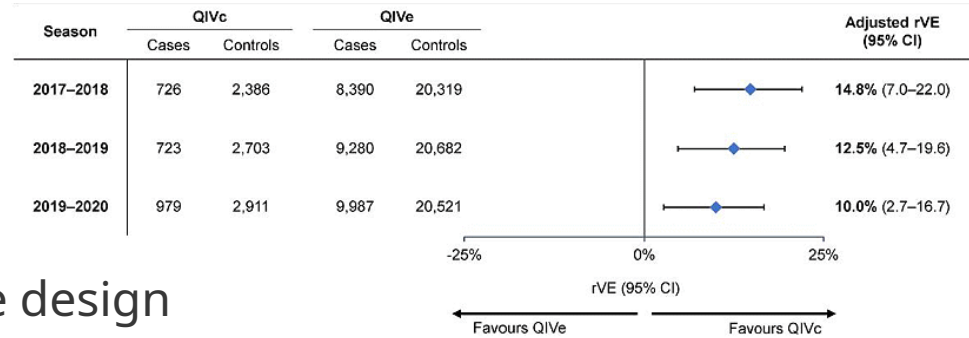
Relative effectiveness of cell-cultured and egg-based influenza vaccines among elderly persons ≥ 65 years in the United States, 2017–2018



資料來源：
Izurieta HS. J Infect Dis 2018; 220: 1255.

Relative effectiveness of cell-cultured and egg-based influenza vaccines

N=31824+33388+34398, 4~64 yrs, 2017~2020, USA

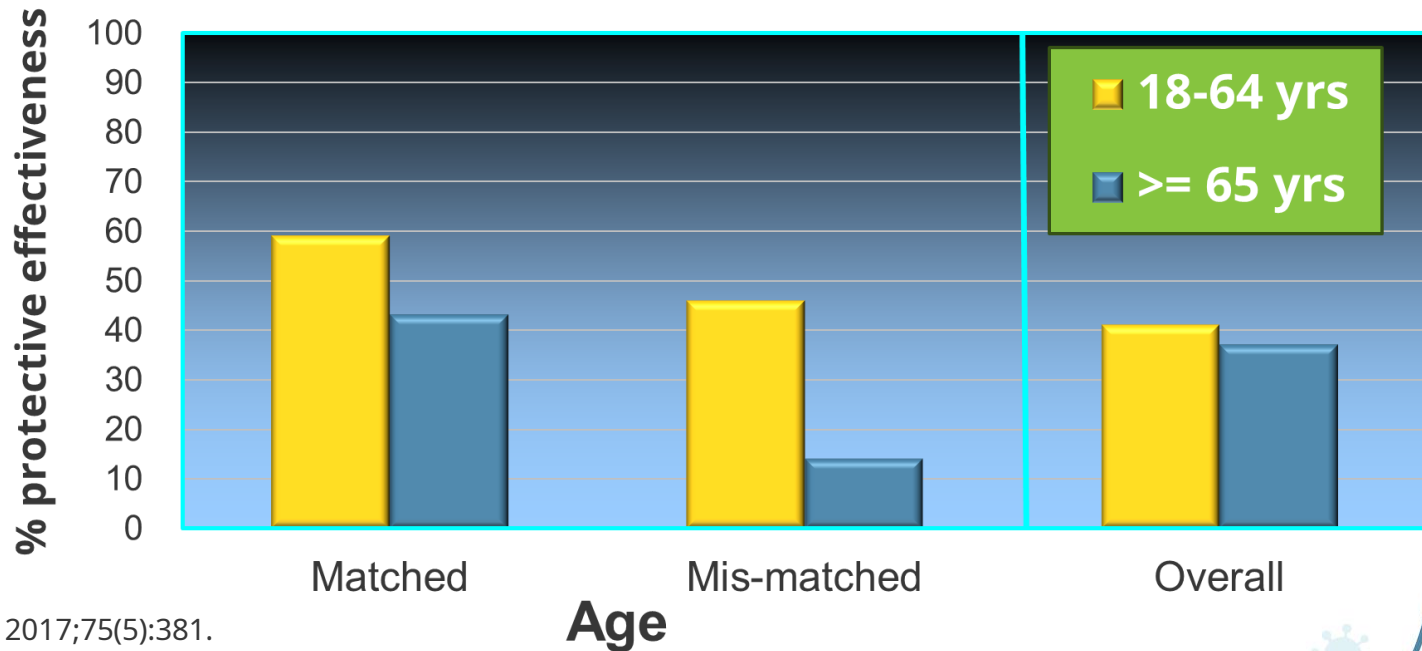


- Retrospective test-negative design
- Relative effectiveness: cell-based vs. egg-based
 - 2017~2018: 14.8% (95% CI, 7.0%–22.0%)
 - 2018~2019: 12.5% (95% CI, 4.7%–19.6%)
 - 2019~2020: 10.0% (95% CI, 2.7%–16.7%)

資料來源：
Stein AN. Open Forum Infect Dis 2024;11(5):ofae175.

長者接種流感的疫苗的保護效果較差

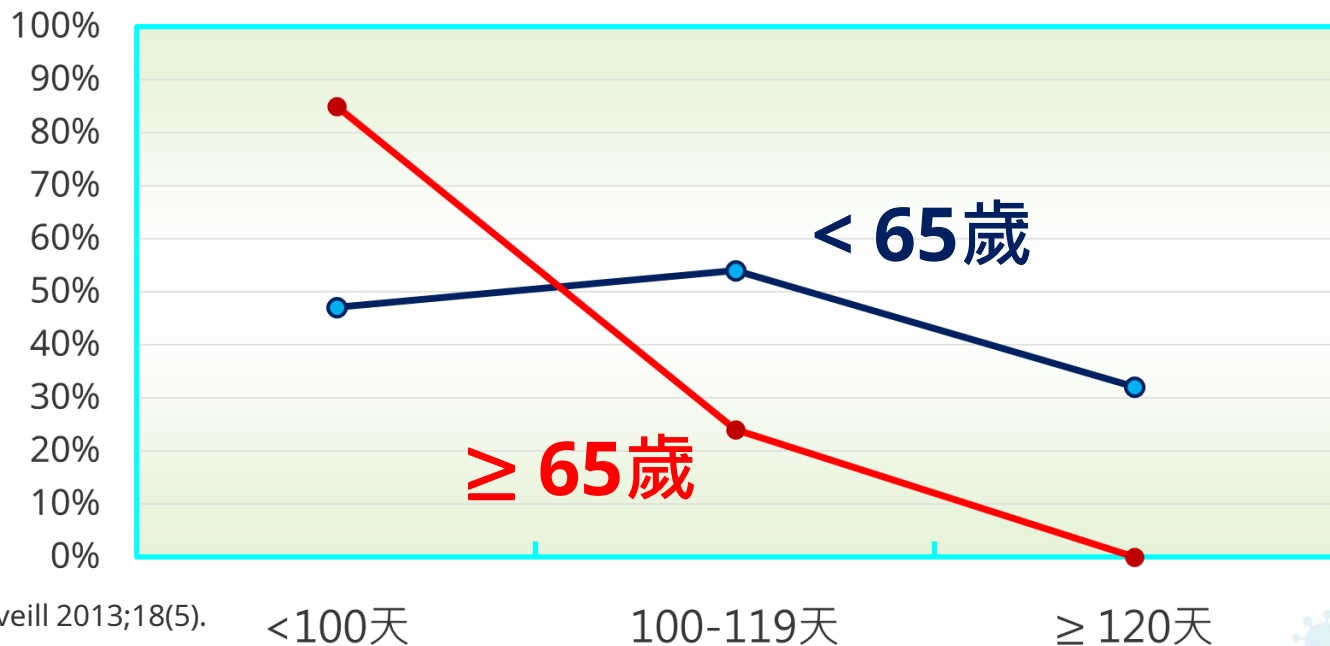
2010~2015，N= 30，meta-analysis



資料來源：Rondy M. J Infect 2017;75(5):381.

65歲以上的流感疫苗保護力快速下降

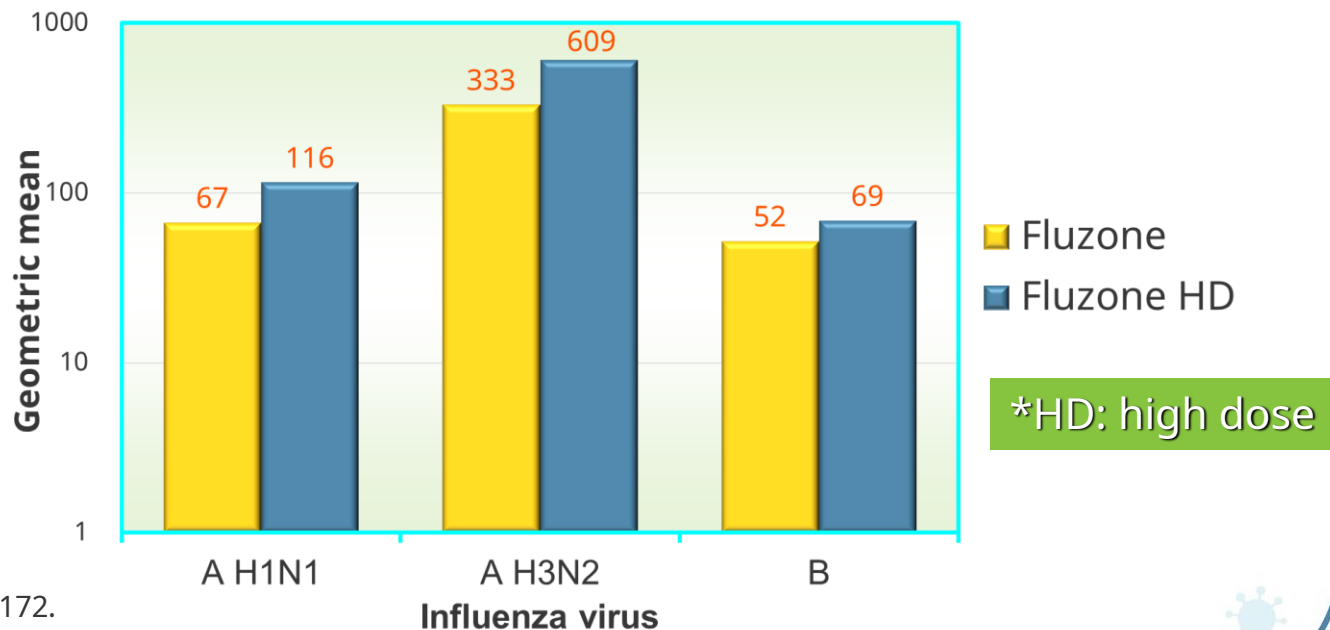
所有年齡，西班牙，2011-2012



資料來源：Castilla J. Euro Surveill 2013;18(5).

Immunogenicity of high-dose vs. standard dose influenza vaccine (65+ years)

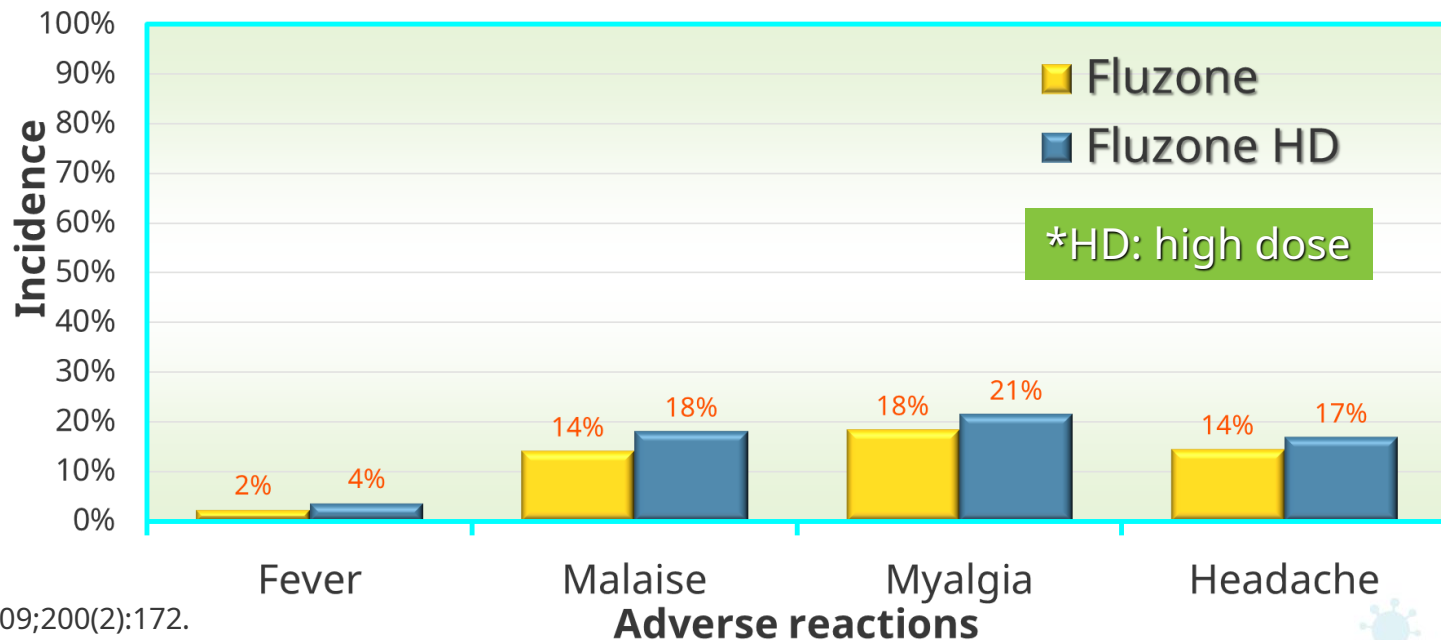
N=2576, 2009, Sanofi-Pasteur



資料來源：
Falsey AR. J Infect Dis 2009;200(2):172.

Safety of high-dose vs. standard dose influenza vaccine (65+ years)

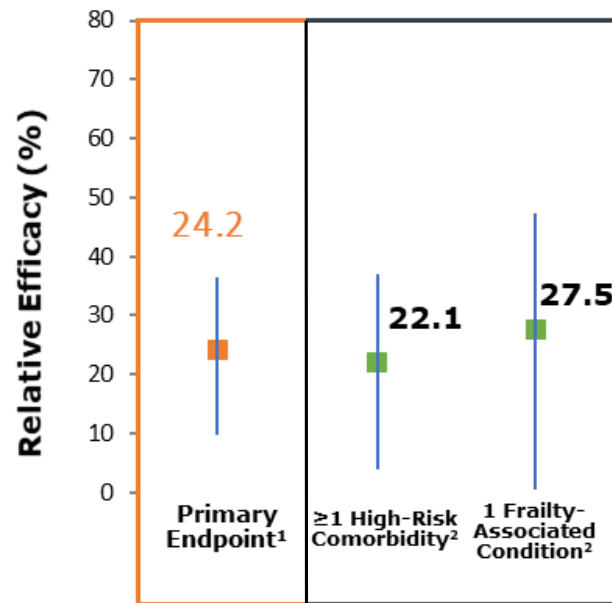
N=2576, 2009, Sanofi-Pasteur



資料來源：
Falsey AR. J Infect Dis 2009;200(2):172.

高劑量疫苗對長者的保護效益

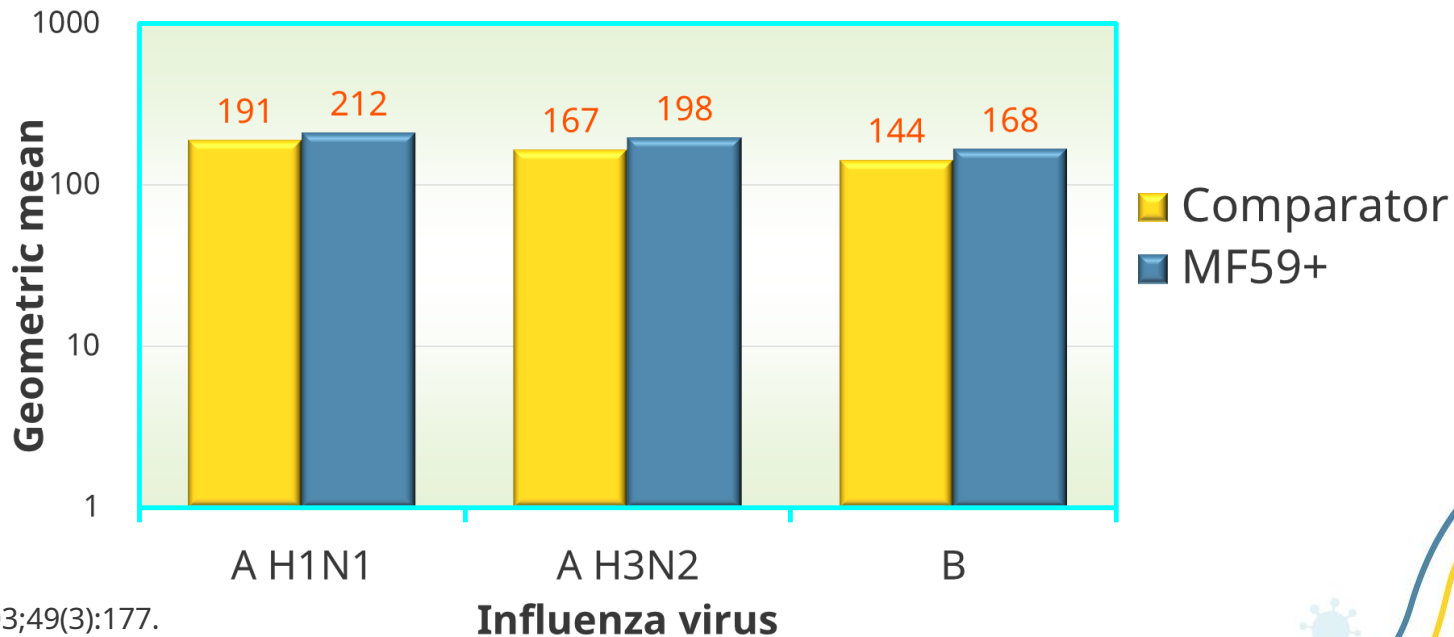
- 高劑量疫苗對長者實驗室確診流感保護效益，相較接種一般流感疫苗提高 24.2%(95% CI 9.7-36.5%)
- 高劑量疫苗對患有1項或沒有高風險共病症長者的保護效益，相較接種一般流感疫苗提高 22.1 (95% CI 3.9%-37.0%)
- 高劑量疫苗對患有1項衰弱相關病症之長者的實驗室確診流感保護效益，相較接種一般流感疫苗提高 27.5%(95% CI 0.4-47.4%)



1. *N Engl J Med* 2014;371:635-45. doi: [10.1056/nejmoa1315727](https://doi.org/10.1056/nejmoa1315727).
2. *Vaccine* 2015;33:4565-71. doi: [10.1016/j.vaccine.2015.07.003](https://doi.org/10.1016/j.vaccine.2015.07.003).

Fluad with MF59 adjuvant in the elderly (65+ years)

N=3272, 13 clinical trial, Europe, USA, 2003



資料來源：
Banzhoff A. Gerontology 2003;49(3):177.

Effectiveness of MF59-Adjuvanted influenza vaccine vs. standard vaccine (elderly)

Review, 2017

- Adjusted risk ratio of MF59-TIV vs. non-adjuvanted vaccines
 - Hospitalizations due to pneumonia/influenza: **0.75**
 - Laboratory-confirmed influenza: **0.37**

Relative effectiveness of MF59-TIV (MF59-TIV versus other vaccines).

Study [Ref.]	Design	Setting	Comparator	Outcome	ES (95% CI)	aES (95% CI)
Iob [25]	Prospective	Long-term care facility	IM-TIV	Influenza-like illness	0.66 (0.53–0.82)^{a,b}	N/A
Mannino [28]	Prospective	Community	IM-TIV	Hospitalization for pneumonia and influenza	0.97 (0.74–1.25) ^c	0.75 (0.57–0.98)^c
Puig-Barberà [31]	Retrospective	Community	Virosomal TIV	Influenza-related hospitalization	0.53 (0.53–1.30) ^c	0.85 (0.54–1.34) ^{c,d}
Puig-Barberà [31]	Retrospective	Community	Virosomal TIV	Laboratory-confirmed influenza	0.72 (0.44–1.18) ^c	0.94 (0.37–2.38) ^{c,e}
Gasparini [30]	Case-control	Community	ID-TIV	Hospitalization for pneumonia and influenza	0.43 (0.20–0.91)^{a,f}	0.75 (0.46–1.24) ^{c,d}
Van Buynder [29]	Prospective case-control	Mixed	IM-TIV	Laboratory-confirmed influenza	0.58 (0.31–1.09) ^c	0.84 (0.31–2.26) ^{c,e}
						0.54 (0.22–1.29)^{a,f}
						0.37 (0.14–0.96)^a

^a Odds ratio.

^b Converted OR (in order to uniform effect sizes, i.e. MF59-TIV vs others); authors' reported OR was 2.16 (95% CI: 1.56–2.98).

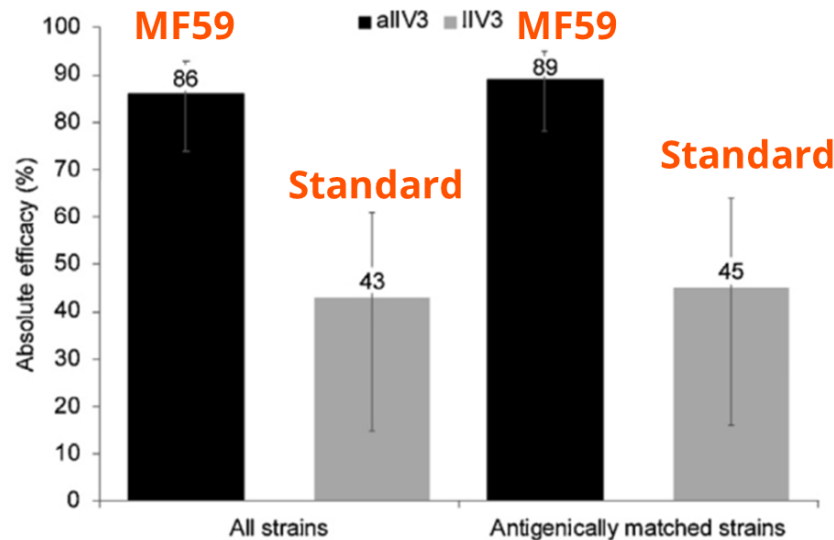
^c Risk ratio.

^d Adjusted model.

^e Multilevel model.

^f Calculated from raw data provided by the corresponding author; statistically significant estimates are in bold.

Effectiveness of MF59-Adjuvanted influenza vaccine vs. standard vaccine (6 mo ~ 5 yrs) 2019

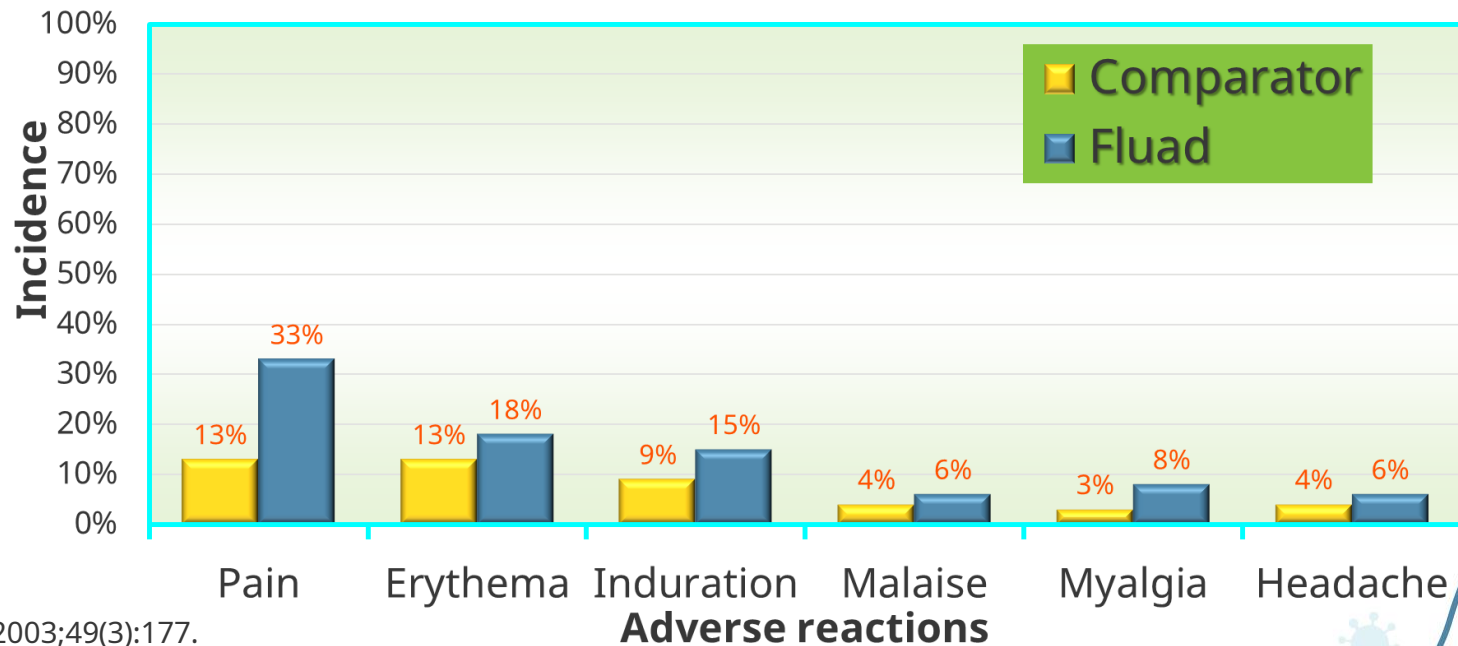


Relative efficacy (%)	75	80
95% CI	55-87	59-90

資料來源：
Lindert K. Int J Infect Dis 2019;85S:S10.

Fluad with MF59 adjuvant in the elderly (65+ years)

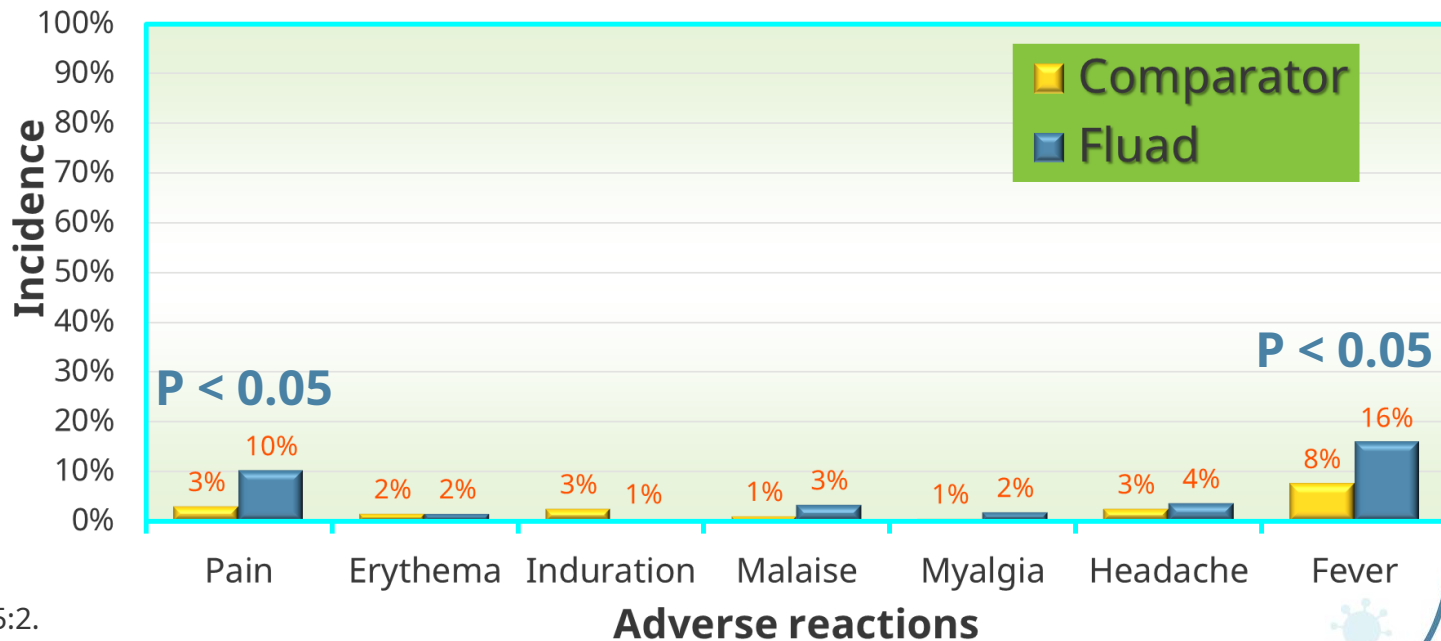
N=3272, 13 clinical trial, Europe, USA, 2003



資料來源：
Banzhoff A. Gerontology 2003;49(3):177.

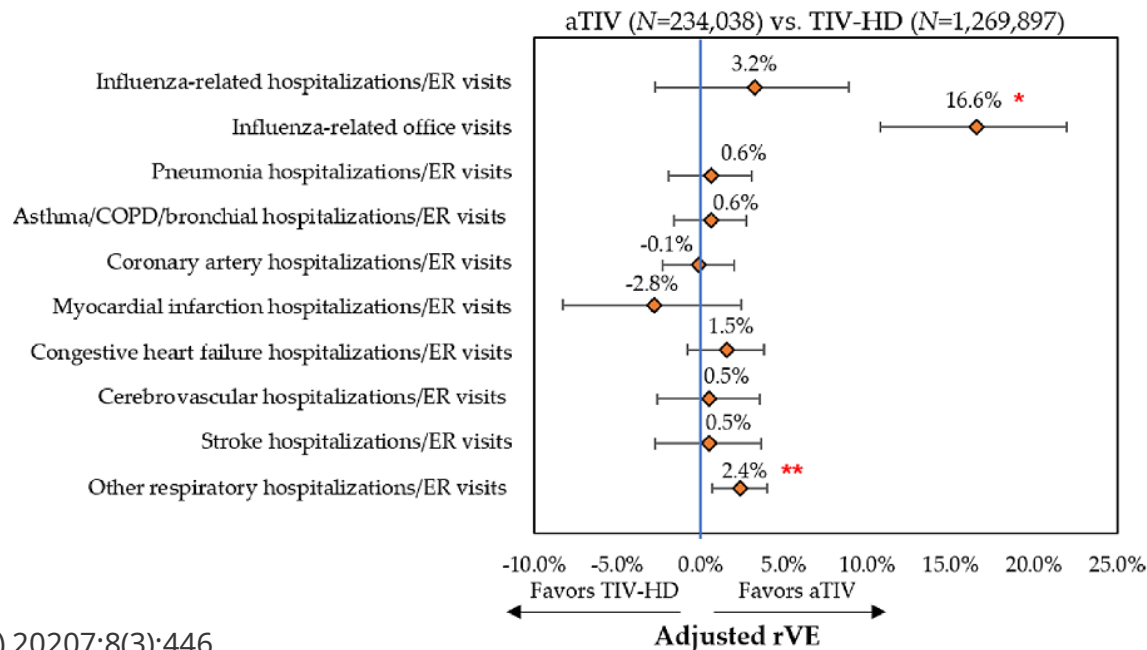
Fluad with MF59 adjuvant in the elderly (60+ years)

N=600, 2008, China



資料來源：
Li R. Immun Ageing 2008;5:2.

Effectiveness of adjuvanted vs. high-dose TIV (65+ years) 2017~2018, USA



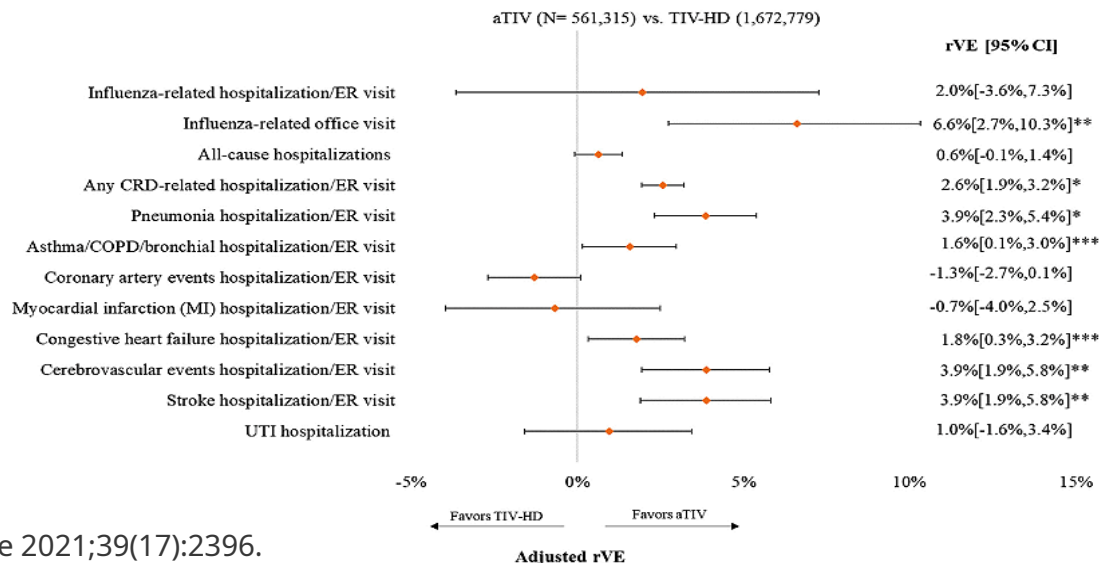
資料來源：Pelton SI. Vaccines (Basel) 2020;7(3):446.

Effectiveness of adjuvanted vs. high-dose TIV (65+ years)

2018~2019, USA

■ Adjuvanted TIV略優於high-dose TIV

**Favor
high-dose TIV**



**Favor
adjuvanted TIV**

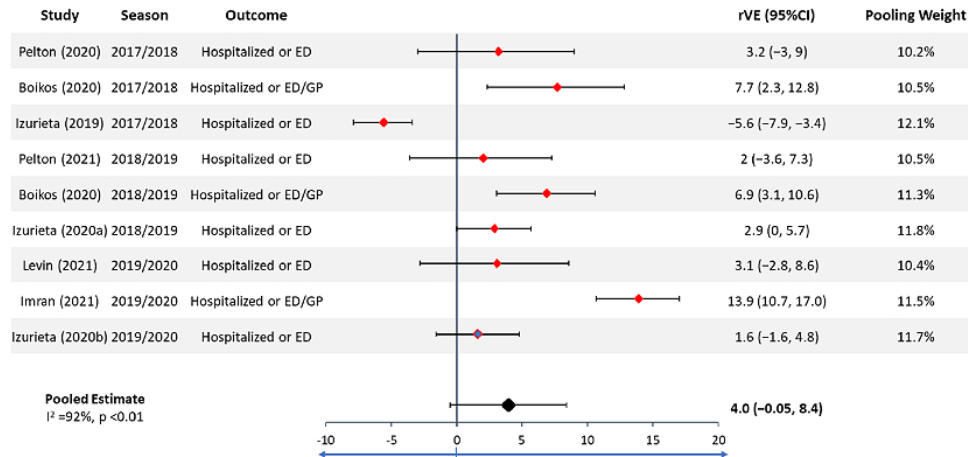
資料來源：Pelton SI. Vaccine 2021;39(17):2396.

Effectiveness of adjuvanted vs. high-dose TIV (65+ years)

Meta-analysis, 2022

■ Meta-analysis: adjuvanted TIV > high-dose TIV

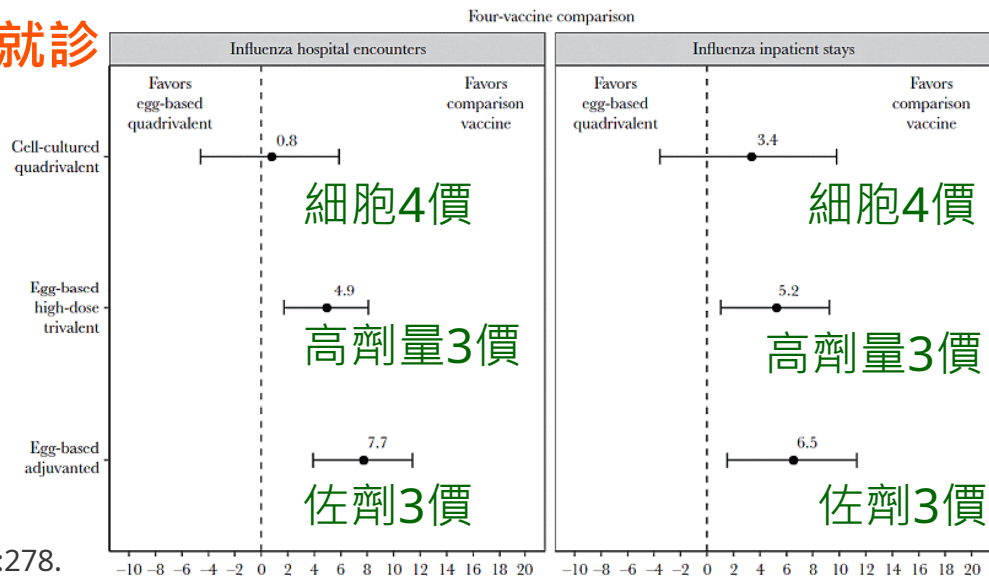
Favor
high-dose TIV



Favor
adjuvanted TIV

Relative effectiveness of influenza vaccines among elderly persons ≥ 65 years 2018-2019, United States

流感就診

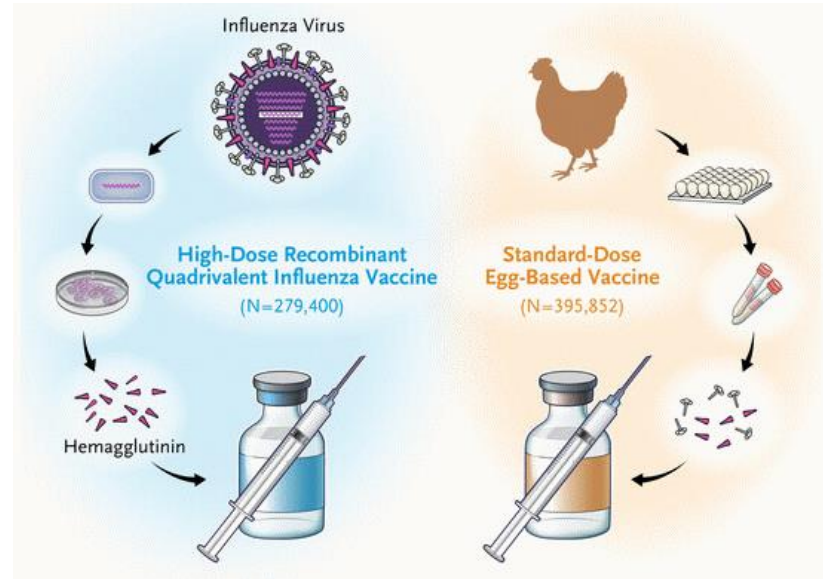


流感住院

資料來源：
Izurieta HS. J Infect Dis 2020;222(2):278.

Recombinant quadrivalent influenza vaccine (Sanofi)

- Manufactured without chicken eggs
- 3X recombinant hemagglutinin protein

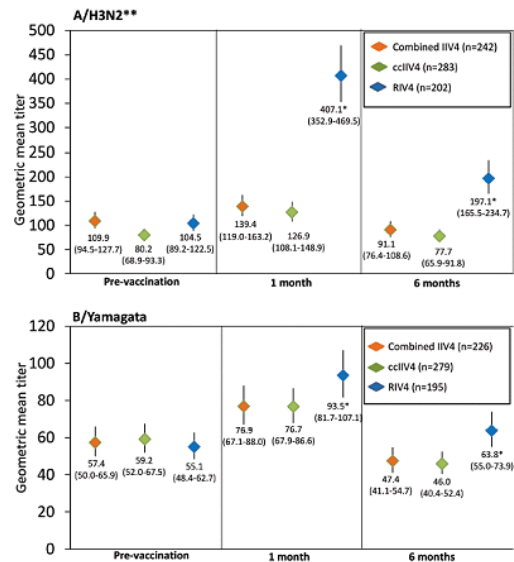
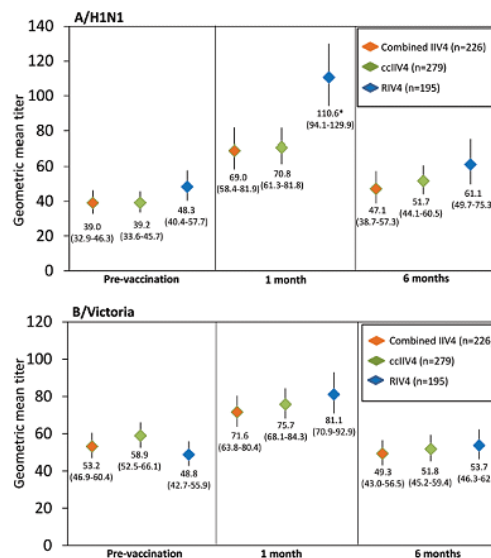


資料來源：
Hsiao A. N Engl J Med 2023;389(24):2245.

Antibody response to 2018–19 cell-based, egg-based, recombinant quadrivalent influenza vaccines

18~64 yrs, United States, 2018~2019

- Immunogenicity:
recombinant > cell-based $\approx$ inactivated vaccine

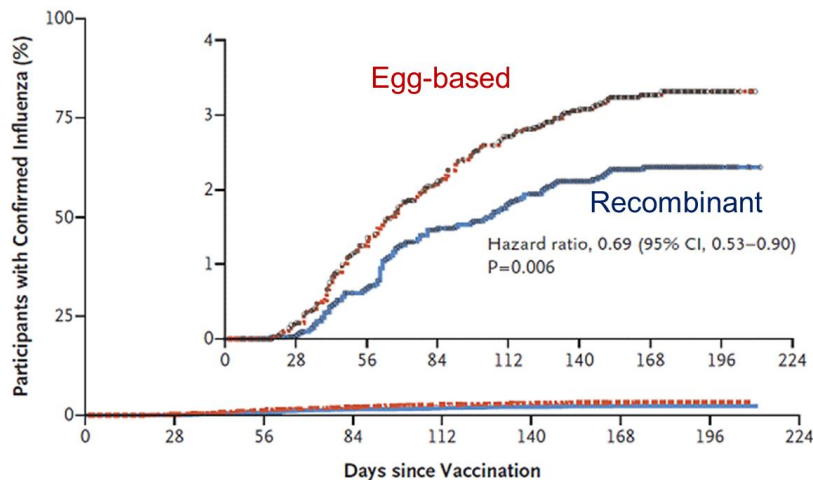


資料來源：
Dawood FS. Clin Infect Dis 2021;73(11):1973.

Recombinant quadrivalent influenza vaccine (Sanofi) vs. egg-based injected influenza vaccine

N=8855, ≥ 50 yrs, 2014~2015, United States

- Recombinant: influenza-like illness ↓ 30% (95%CI 10%-47%)



資料來源：
Dunkle LM. N Engl J Med 2017;376:2427.

Recombinant quadrivalent influenza vaccine (Sanofi) vs. egg-based injected influenza vaccine

N=1630328, < 65 yrs, 2018~2020, United States

■ Recombinant/egg-based, relative vaccine effectiveness

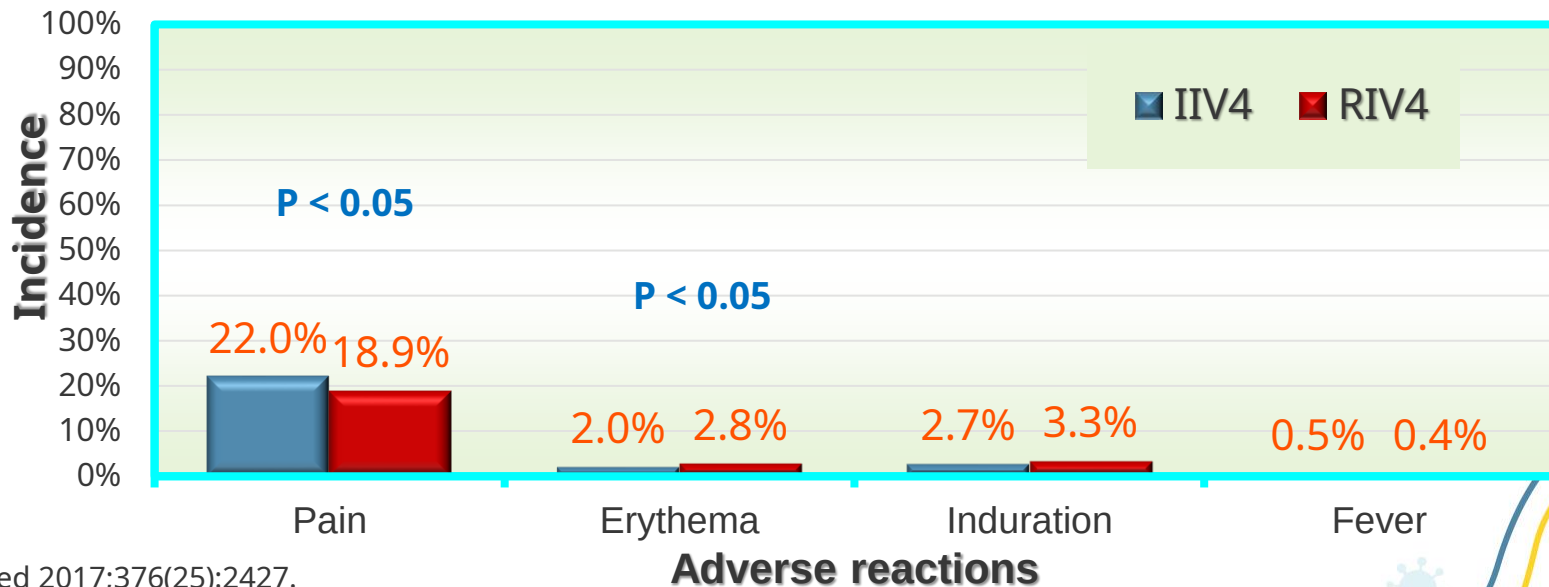
- 50~64 yrs: 15.3%
(95%CI 6.0%-24.5%)
- 18~49 yrs: 10.8% (95%CI 6.6%~14.7%)
- No difference in hospitalization

Table 2. Primary and Secondary Outcomes and Relative Vaccine Effectiveness among Participants 50 to 64 Years of Age.*

Outcome	Recombinant Vaccine (N = 279,400)	Standard-Dose Vaccine (N = 395,852)	Unadjusted Rate Ratio	Adjusted Hazard Ratio (95% CI)†	Relative Vaccine Effectiveness (95% CI)	P Value‡
	no. of cases per 1000				%	
Primary outcome						
PCR-confirmed influenza	559 (2.00)	925 (2.34)	0.86	0.85 (0.76 to 0.94)	15.3 (5.9 to 23.8)	0.002
Secondary outcomes						
PCR-confirmed influenza A	522 (1.87)	862 (2.18)	0.86	0.84 (0.76 to 0.94)	15.7 (6.0 to 24.5)	0.002
PCR-confirmed influenza B	37 (0.13)	64 (0.16)	0.82	0.90 (0.60 to 1.34)	10.3 (−33.9 to 39.9)	0.59
Hospitalization for PCR-confirmed influenza	95 (0.34)	153 (0.39)	0.88	0.84 (0.65 to 1.09)	15.9 (−9.2 to 35.2)	0.19
Hospitalization for community-acquired pneumonia	106 (0.38)	183 (0.46)	0.82	0.83 (0.66 to 1.06)	16.7 (−5.6 to 34.4)	0.13
Hospitalization for cardiorespiratory event	631 (2.26)	890 (2.25)	1.004	0.98 (0.88 to 1.08)	2.4 (−8.1 to 11.9)	0.64

資料來源：
Dunkle LM. N Engl J Med 2017;376:2427.

Safety of recombinant vs. egg-based QIV in the elderly (50+ years) 2014 ~ 2015, USA



資料來源：
Dunkle LM. N Engl J Med 2017;376(25):2427.

Relative effectiveness of influenza vaccines among elderly persons ≥ 65 years

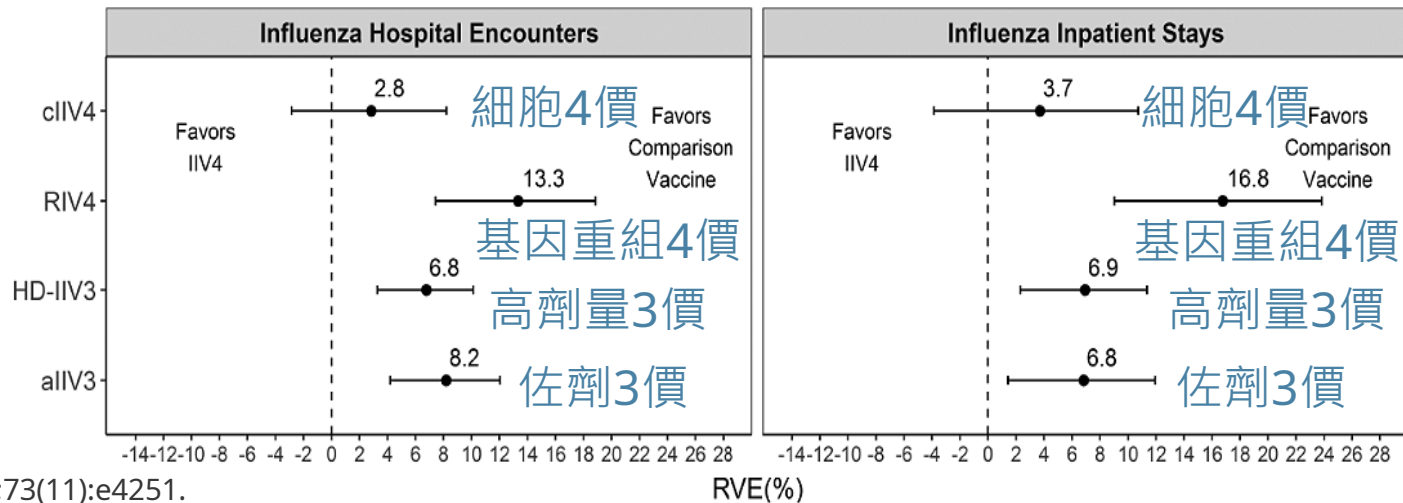
2019-2020, United States



流感就診

Five-Vaccine Comparison

流感住院



資料來源：
Izurieta HS. Clin Infect Dis 2021;73(11):e4251.



Prevention and control of seasonal influenza with vaccines

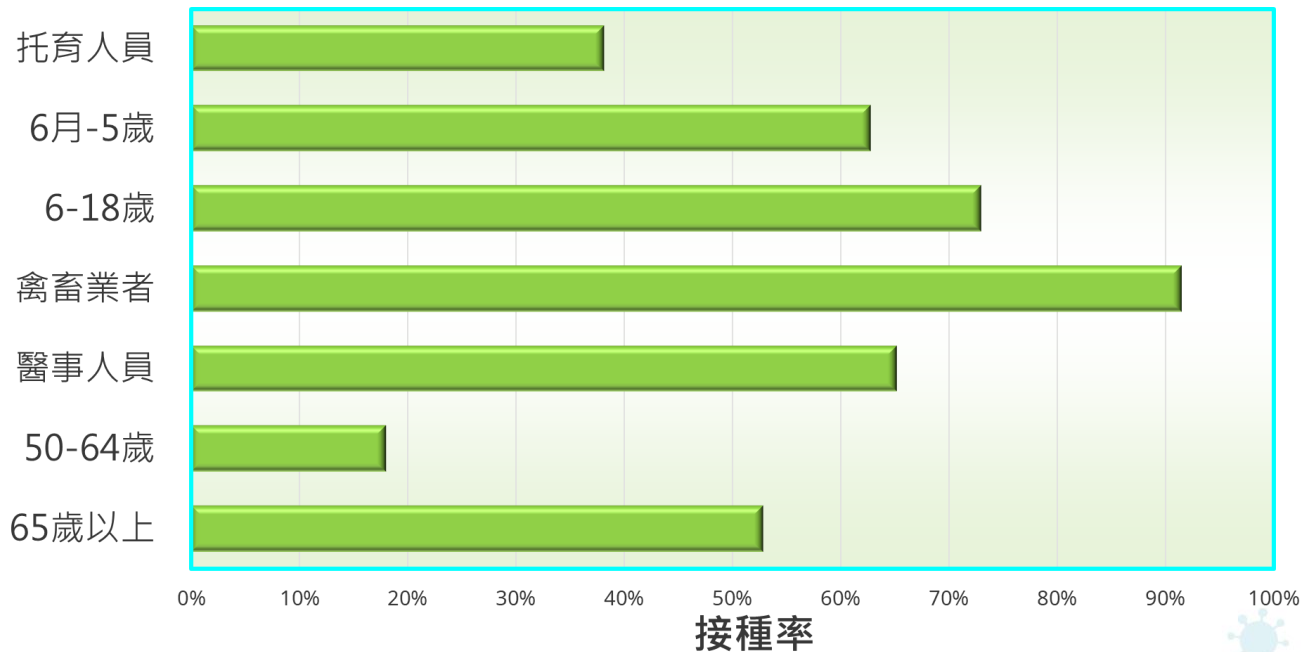
ACIP, USA, 2022

- ACIP recommends that adults aged ≥ 65 years preferentially receive any one of the following higher dose or adjuvanted influenza vaccines:

- Quadrivalent high-dose inactivated influenza vaccine (HD-IIV4) : HA 60 μg
- Quadrivalent recombinant influenza vaccine (RIV4): HA 45 μg
- Quadrivalent adjuvanted inactivated influenza vaccine (aIIV4): HA 15 μg
- HD-IIV, RIV, aIIV: similar effectiveness

資料來源 : Grohskopf LA. MMWR 2022;71(1):1.

2023~2024年 臺灣公費流感疫苗接種率



資料來源：疾病管制署資料

2023~2024.1.15

醫事人員流感疫苗接種率

醫事類別	醫事人員執登數	醫事人員接種數	醫事人員接種率
醫院	195,604	155,108	79.30%
診所	84,285	45,039	53.44%
其他	85,554	44,314	51.80%
-藥師/藥劑生自營/駐店管理	18,857	6,762	35.86%
-學校醫務室	6,918	4,483	64.80%
-眼鏡公司(商號)合設	6,825	973	14.26%
-心理諮商/治療所	2,820	704	24.96%
-事業單位醫務室	1,910	718	37.59%
-牙體技術所	2,435	305	12.53%
-醫事檢驗所	1,046	255	24.38%
-捐血中心	637	377	59.18%
總計	365,443	244,461	66.89%

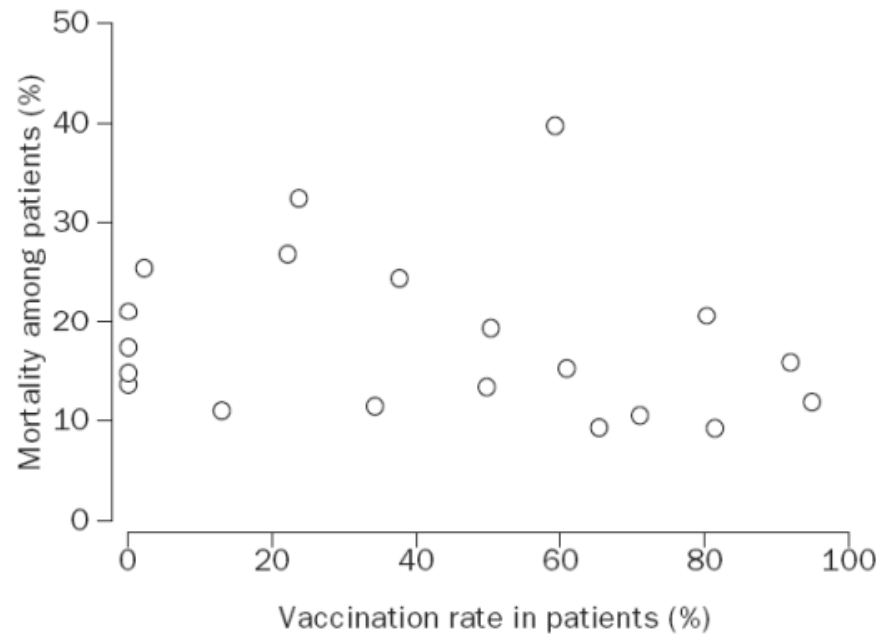
資料來源：疾病管制署資料

健康照護者接種流感疫苗對慢性 照護老人死亡率的影響

UK, 1996-7, 20 elderly-care hospitals



- High vaccine vs. low vaccine coverage: mortality ↓ 42%



資料來源：Carman WF. Lancet 2000;355(9198): 93.



Influenza Vaccination of Health-Care Personnel

Healthcare Infection Control Practices Advisory Committee (HICPAC) and the Advisory Committee on Immunization Practices (ACIP), 2006

- 教育健康照護人員流感疫苗的益處
- 每年提供免費流感疫苗可保護健康照護人員、病人與家人
- 採用可提高流感疫苗接種率的措施，例如疫苗門診、可動推車
- 無禁忌而拒絕接種疫苗者必須簽署拒絕文件
- 定期追蹤疫苗接種率
- 以流感接種率做為病人安全品質計畫的指標之一

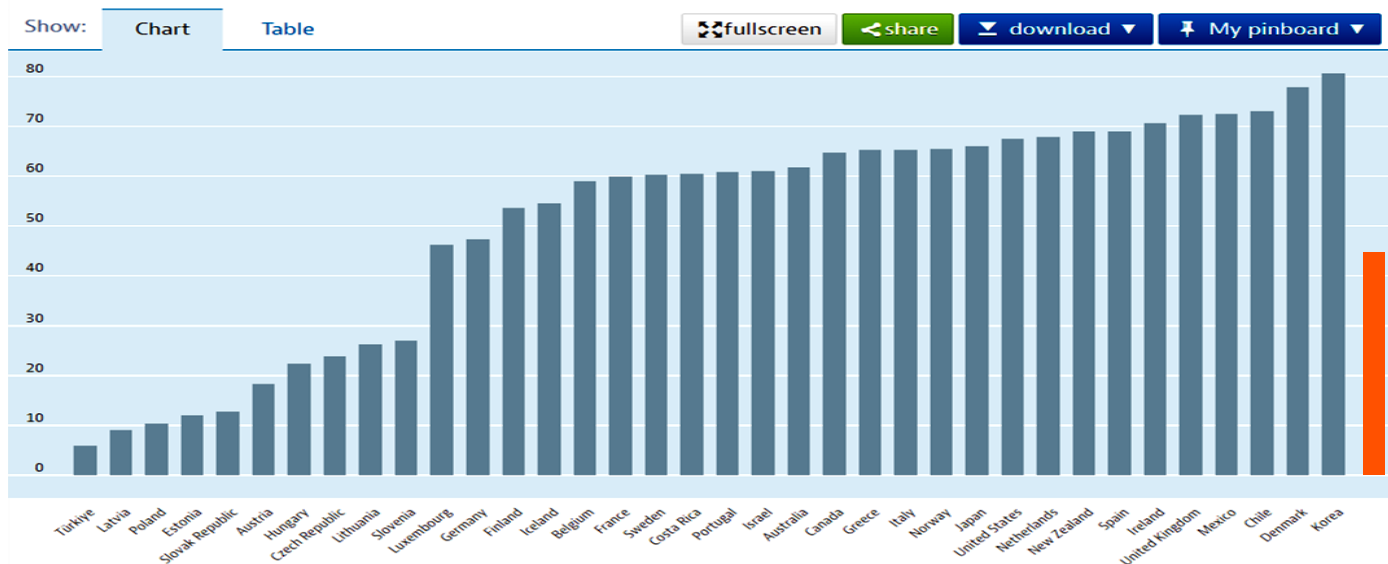
長者流感疫苗涵蓋率 (OECD)

Influenza vaccination rates in OECD countries



Influenza vaccination rates Total, % of population aged 65+, 2021 or latest available

Source: Health care utilisation



資料來源：

<https://data.oecd.org/healthcare/influenza-vaccination-rates.htm>, slide kindly provided by Dr. Chen PY.



不同族群新冠疫苗猶豫者主要的原因

2022.1.21，報導者

■ 年長者

- 擔心年齡與身體狀況無法承受疫苗副作用
- 接收訊息管道單一、對政治不信任
- 認為年輕人風險高應先接種

■ 慢性病患

- 擔心難以承受疫苗的副作用，在染疫風險與疫苗副作用中掙扎
- 沒有明確的疫苗適用指引
- 擔心疾病隱私曝光

不同族群新冠疫苗猶豫者主要的原因

2022.1.21，報導者

■ 孕婦

- 政府對孕婦、胎兒的**接種風險說明**太少
- 國內缺乏孕婦接種的統計數據、後續追蹤資料
- 態度保守，傾向信任親友或婦產科醫師

■ 藍領與弱勢族群

- 對疫苗認識有限
- 脆弱生活情境，**擔心難以承受疫苗副作用**
- 對政府、體制長期不信任

新冠、流感、肺炎鏈球菌同時來... 民眾身體能一次承受這麼多疫苗嗎？

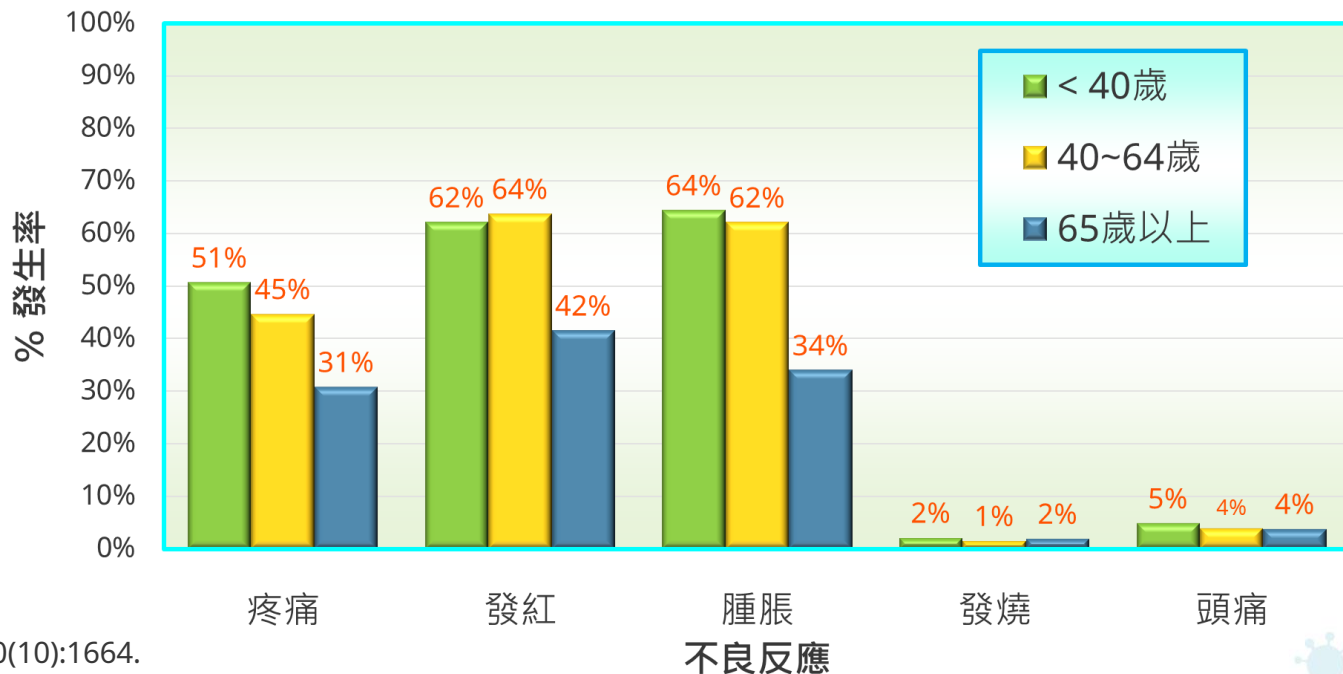
2021.10.1 信傳媒

- 若在身體狀況不是很好的時候，比如在發炎反應當中，或是其他小感冒時要先避開，「因為新冠疫苗的確有一些人會有較強的反應，像年長者的確是需要被疫苗保護的族群，但長輩的情況有時也不太穩定，還是建議等身體狀況較OK的時候再施打，較能夠承受住一些副作用。」
- 她也提到，有些長輩打疫苗可能會出現皮膚紅腫的現象，「如果年紀真的很大了，皮膚紅腫又會覺得很辛苦，就應該要避開一下，等狀況比較穩定再施打。」

資料來源：<https://www.cmmedia.com.tw/home/articles/30086>

接種流感疫苗的不良反應

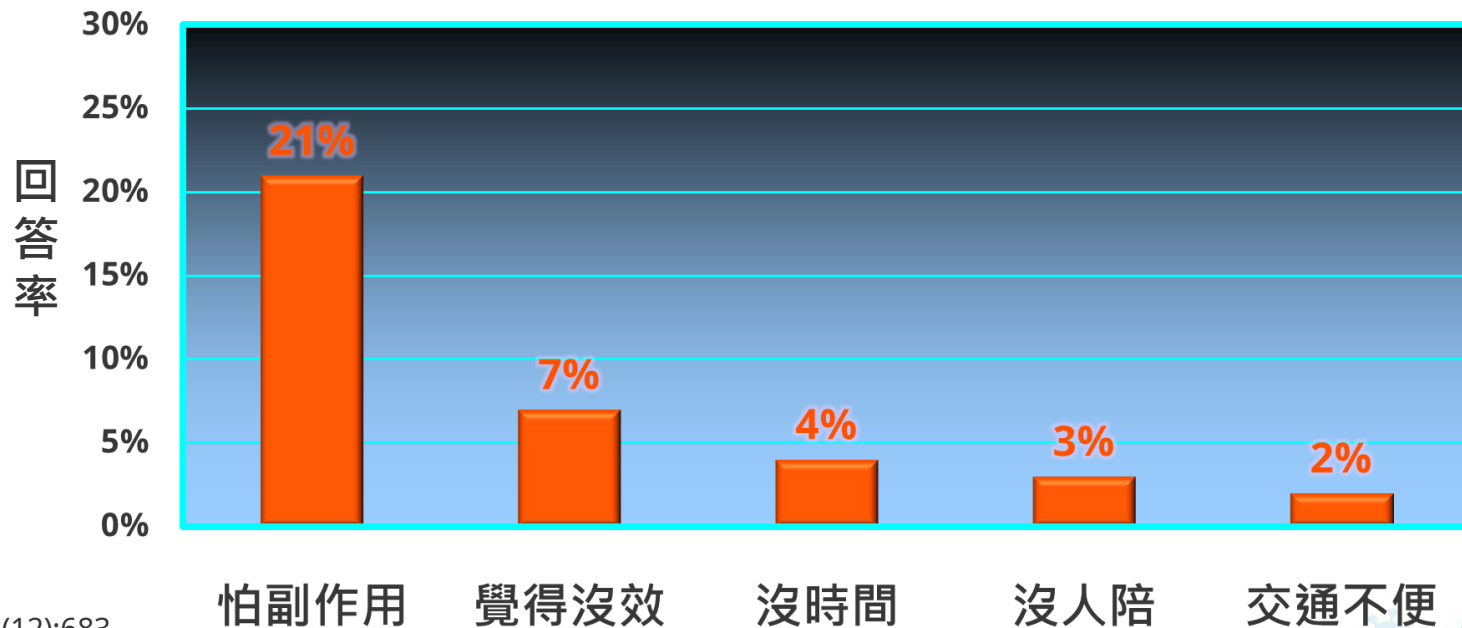
N=1493，2019~2020，日本



資料來源：
Kumabe A. Vaccines. 2022; 10(10):1664.

免費流感疫苗的障礙因素

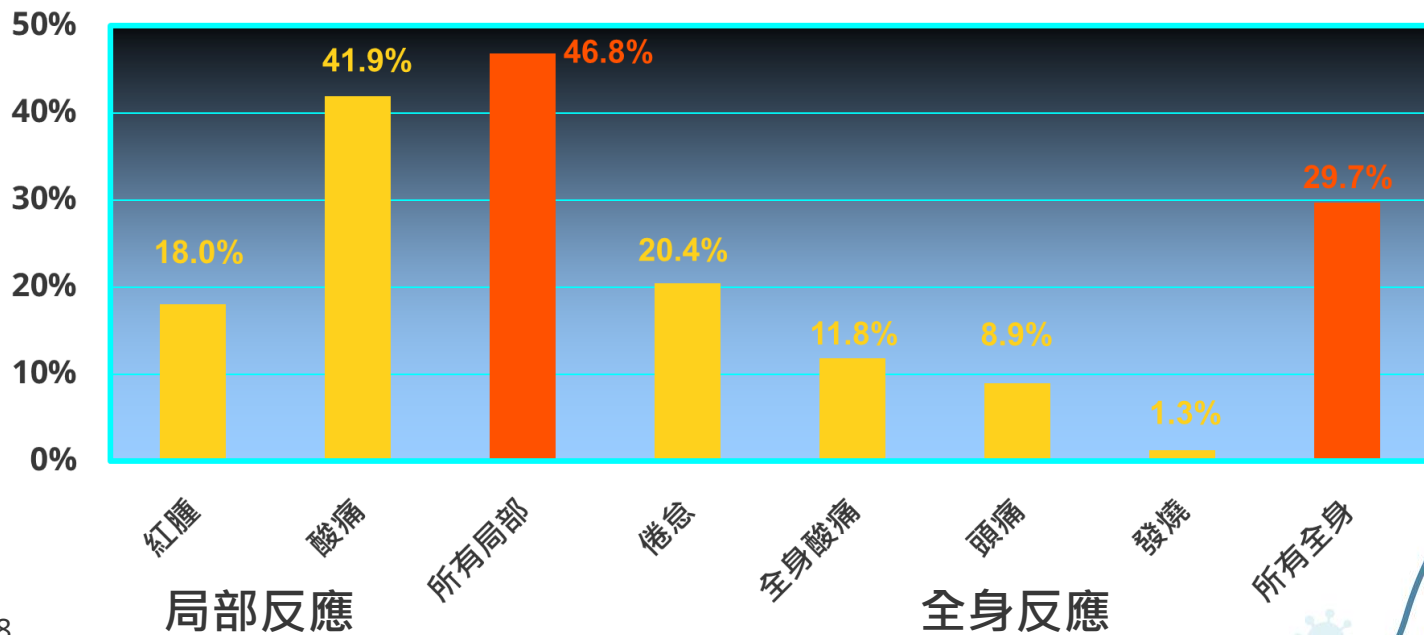
2007年10月 65歲以上老人調查，N=1,115



資料來源：
黃惠萍. 疫情報導 2007;23(12):683.

流感疫苗副作用 (Fluarix)

2004 全臺灣醫護人員問卷調查 N=1785



資料來源：
蕭惠心. 疫情報導 2005; 21: 618.

流感疫苗的禁忌與注意事項

疾病管制署，2018



■ 禁忌

- 先前接種本疫苗或對本疫苗任何成分曾發生嚴重過敏反應者

■ 注意事項

- 發燒或正患有急性中重度疾病者，宜待病情穩定後再接種
- 已知對「蛋」之蛋白質有嚴重過敏者，可在門/住診由熟悉處理過敏症狀之醫事人員提供接種，並於接種後觀察30 分鐘，無不適症狀再離開

資料來源：

<http://www.cdc.gov.tw/page.aspx?treeid=D78DE698C2E70A89&nowtreeid=035002D5848EF51F>, access June 10, 2014



流感疫苗的禁忌與注意事項

疾病管制署，2018



■ 注意事項

- 出生未滿6個月
- 先前接種本疫苗6週內曾發生Guillain-Barré 症候群 (多發性神經炎) 者
- 其他經醫師評估不適合接種者



資料來源：

<http://www.cdc.gov.tw/page.aspx?treeid=D78DE698C2E70A89&nowtreeid=035002D5848EF51F>, access June 10, 2014



立即型過敏

Anaphylaxis



- Incidence: 0.65 – 1.53 cases/million doses of vaccine
- Vaccines: measles-mumps-rubella, hepatitis B, diphtheria-tetanus, diphtheria-tetanus-pertussis, Haemophilus influenzae type b, oral polio vaccines.
- Providers should be prepared to provide immediate medical treatment should it occur. (Red Book)
- 2004.2.3 流感防治諮詢委員會：接種流感疫苗後，必須於原醫療站休息至少30分鐘

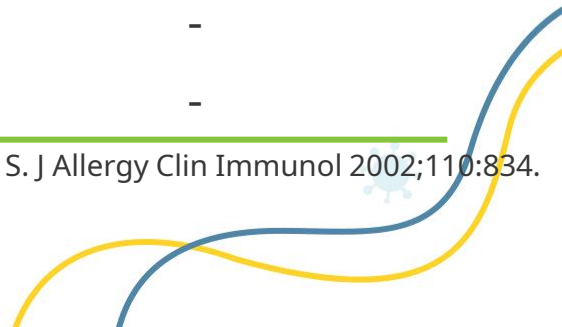


雞蛋過敏者接種非活性 流感疫苗



研究者	個案數	副作用比率	症狀
Ratner, 1952	5	40%	蕁麻疹、wheeze
Miller, 1983	3	33%	癢、咽部症狀
Murphy, 1985	6	17%	wheeze
Zeiger	28	7%	聲音沙啞、咳嗽 鼻水、蕁麻疹
Anolik, 1992	8	0	-
James, 1998	83	0	-

資料來源：Robert S. J Allergy Clin Immunol 2002;110:834.



蛋過敏者接種流感疫苗

衛生福利部傳染病防治諮詢會



- 106年11月16日衛生福利部傳染病防治諮詢會流感防治組、預防接種組聯席會議會議決議：將「已知對『蛋』之蛋白質有嚴重過敏者」自接種禁忌症移除，惟應於注意事項(precaution)加列對蛋嚴重過敏者接種疫苗之相關說明內容
- 接種須知：已知對「蛋」之蛋白質有嚴重過敏者，可在門/住診由熟悉處理過敏症狀之醫事人員提供接種，並於接種後觀察30分鐘，無不適症狀再離開



蛋過敏者接種流感疫苗

衛生福利部傳染病防治諮詢會



- 112年8月10日衛生福利部傳染病防治諮詢會流感防治組、預防接種組聯席會議會議決議：刪除上開接種須知中，有關蛋過敏者接種流感疫苗之相關處置文字

蛋過敏者可安心接種流感疫苗



Guillain-Barré syndrome and influenza vaccine

ACIP, CDC, 2008

- The estimated risk for GBS is low (i.e., approximately one additional case per 1 million persons vaccinated). The potential benefits of influenza vaccination in preventing serious illness, hospitalization, and death substantially outweigh these estimates of risk for vaccine-associated GBS.

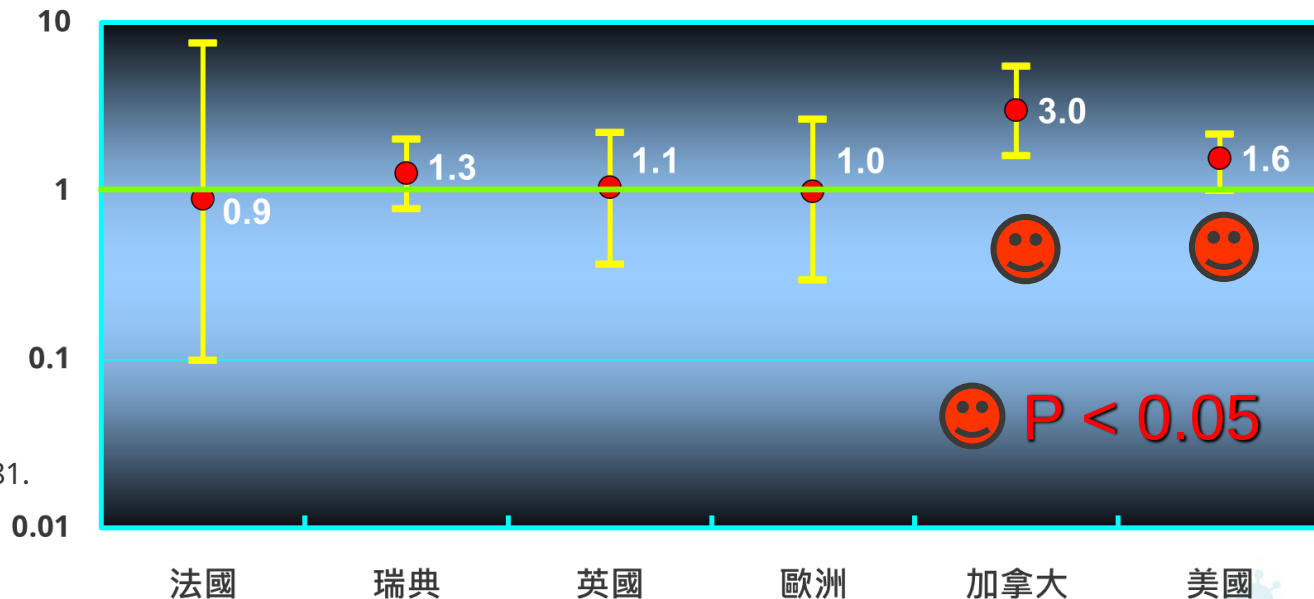
資料來源：

Advisory Committee on Immunization Practices. MMWR 2008;57:1-60.

Guillain-Barré syndrome associated with pandemic H1N1 2009 vaccines



Odds ratio



資料來源：

Grimaldi-Bensouda L. Am J Epidemiol 2011;174(3):326.
Bardage C. BMJ 2011;343:d3908.
Andrews N. Vaccine 2011;29(45):7878.
Dieleman J. BMJ 2011;343:d3908.
De Wals P. JAMA 2012;308(2):175-81.
Wise ME. Am J Epidemiol 2012;175(11):1110.



Gullain-Barré syndrome after trivalent influenza vaccination in Taiwan

2007~2015，健保資料庫

TABLE 4 | Odds ratio of history of trivalent influenza vaccination on GBS patients and control subjects (using conditional logistic regression).

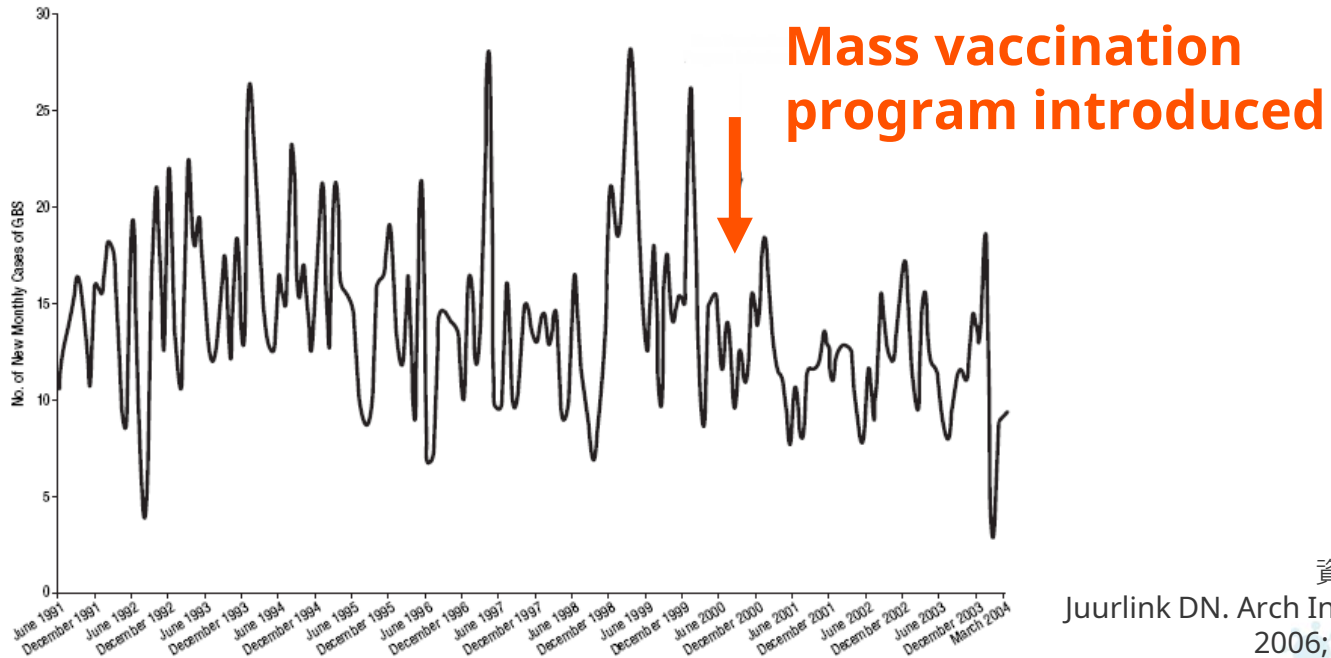
	GBS patients (n = 182)	Control subjects (n = 910)	Odds ratio (95% CI)	p value
	No. (%)	No. (%)		
Flu vaccine within 90 days	17 (9.34)	72 (7.91)	1.26 (0.67–2.38)	0.47
Flu vaccine within 42 days	7 (3.85)	26 (2.86)	1.46 (0.56–3.78)	0.44

CI, confidence interval.

資料來源：
Chang KH. Front Neurol 2019;10:768.

Guillain-Barré syndrome and influenza vaccine

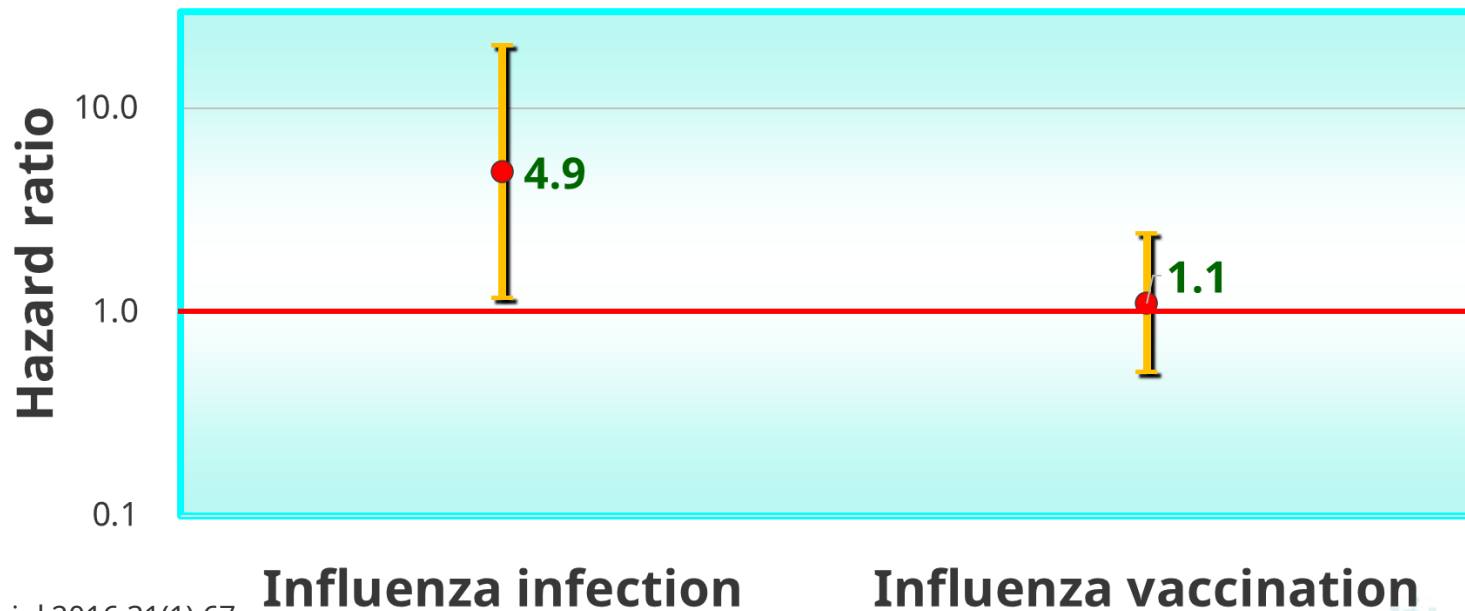
Canada, 1992-2004, N=1,601



資料來源：
Juurlink DN. Arch Intern Med
2006;166:2217.

Risk of Guillain-Barre' syndrome after exposure to pandemic influenza A(H1N1)pdm09 vaccination or infection

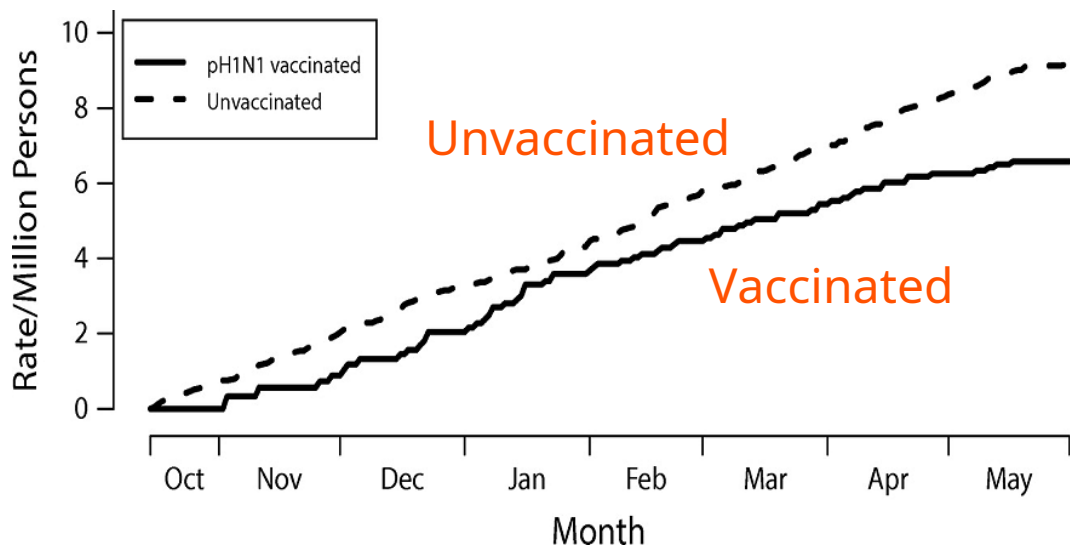
N = 4,832,211, 2015, Norway



資料來源：
Ghaderi S. Eur J Epidemiol 2016;31(1):67.

Cumulative risk of Guillain-Barré syndrome (GBS) among the 2009 pH1N1 vaccinated and unvaccinated groups

15 October 2009 ~ 31 May 2010, USA



資料來源：
Vellozzi C. Clin Infect Dis
2014;58(8):1149.

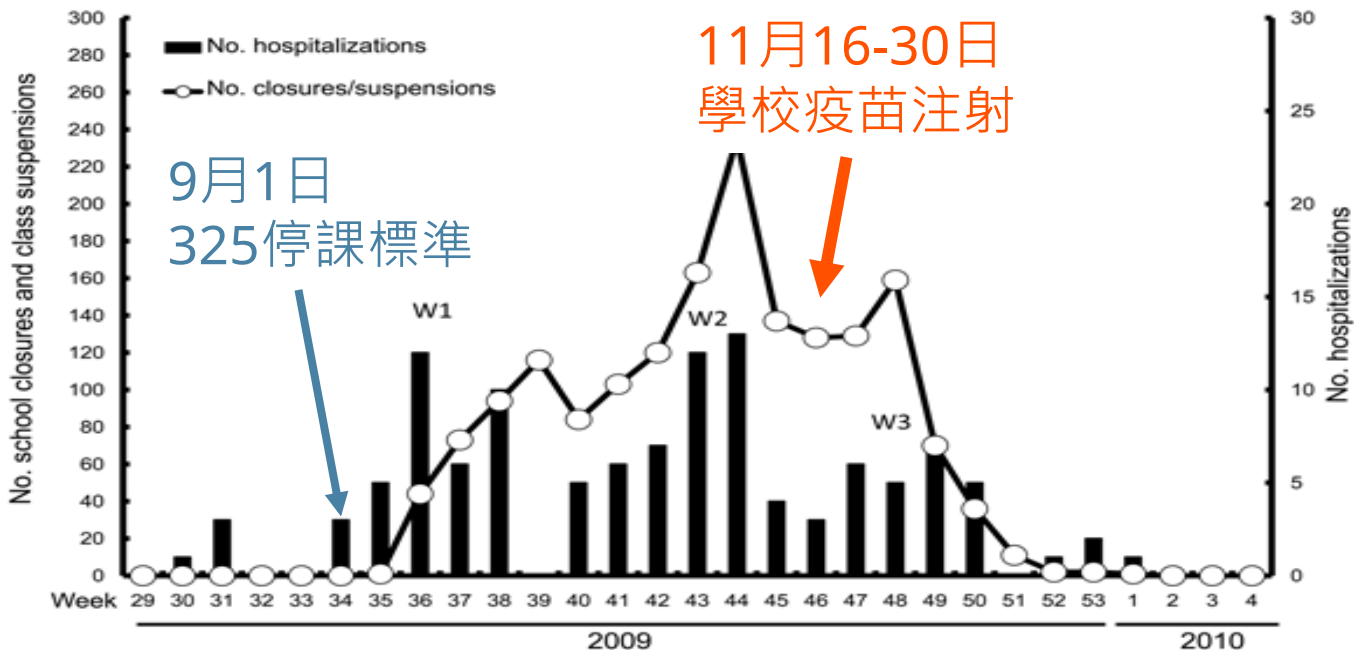
大規模疫苗接種引起的安全性質疑

臺灣

- 施打H1N1疫苗身體抽搐 高三女無法練琴 (今日，2009.12.19)
- 國一女打完疫苗 四肢無力! (華視，2009.12.22)
- 打完疫苗！18歲女半癱 現又吐血眼矇 (TVBS，2009.12.23)
- 孕婦死胎.失聰 疫苗問題連爆 (民視，2009.12.24)
- 劉童死亡效應 投訴暴增 (自由，2009.12.23)
- 小五男童突然腦中風 家屬疑疫苗所害 (今日，2010.1.2)
- 疫苗又惹禍？69歲男子昏迷不醒 (中廣，2010.1.3)

2009 H1N1 新型流感

2009, 臺北市



資料來源：
Hseuh PR. Emerg Infect Dis
2010;16(8):1309.



2009大流行H1N1流感疫苗傷害救濟申訴

案件 N=543

診斷	個案數	無法排除相關%	救濟%
心因性疾病	49	0	
焦慮反應	18	0	22%
轉化症	31	0	55%
皮疹	52	48%	58%
顏面神經麻痺	33	0	97%
腦中風	29	0	3%
周邊神經麻痺	27	30%	52%
流產/胎兒死亡	23	0	70%
習慣性抽動 (tics)	15	0	0
血小板低下	14	13%	100%
腦脊髓炎	14	50%	79%
Guillain-Barré症候群	3	100%	100%

法藥廠流感疫苗奪命 暫停施打

2006.10.24 人間福報



- 以色列南部水牛城，這幾天傳出有**四名五十三歲到七十五歲的糖尿病、心臟病男性病患**，在施打賽諾菲安萬特生產的**流感疫苗後暴斃**，雖然該國衛生部官員不認為他們的死因和注射疫苗有直接關係，但仍下令全國暫停施打該藥廠生產的疫苗，以防萬一
- 消息傳至臺灣，我國衛生署疾管局緊急宣布**暫停施打該藥廠的公費流感疫苗**。根據統計，我國今年向三家藥廠共採購二百二十四萬五千劑流感疫苗，以供應免費接種之用，其中向賽諾菲安萬特訂購的疫苗有一百一十二萬八千劑



流感疫苗可能致死？

2007年，以色列

- 2006年10月四位以色列老人猝死
- 2007 以色列研究：
 - 2003-2004: 25人 (0.03%)
 - 2004-2005: 38人 (0.03%)
 - 2005-2006: 40人 (0.03%)
 - 2006-2007: 29人 (0.03%)



資料來源：
Kokia ES. Vaccine 2007;25:8557.

【國際瞭望】流感疫苗針要命 每年奪走數十命！

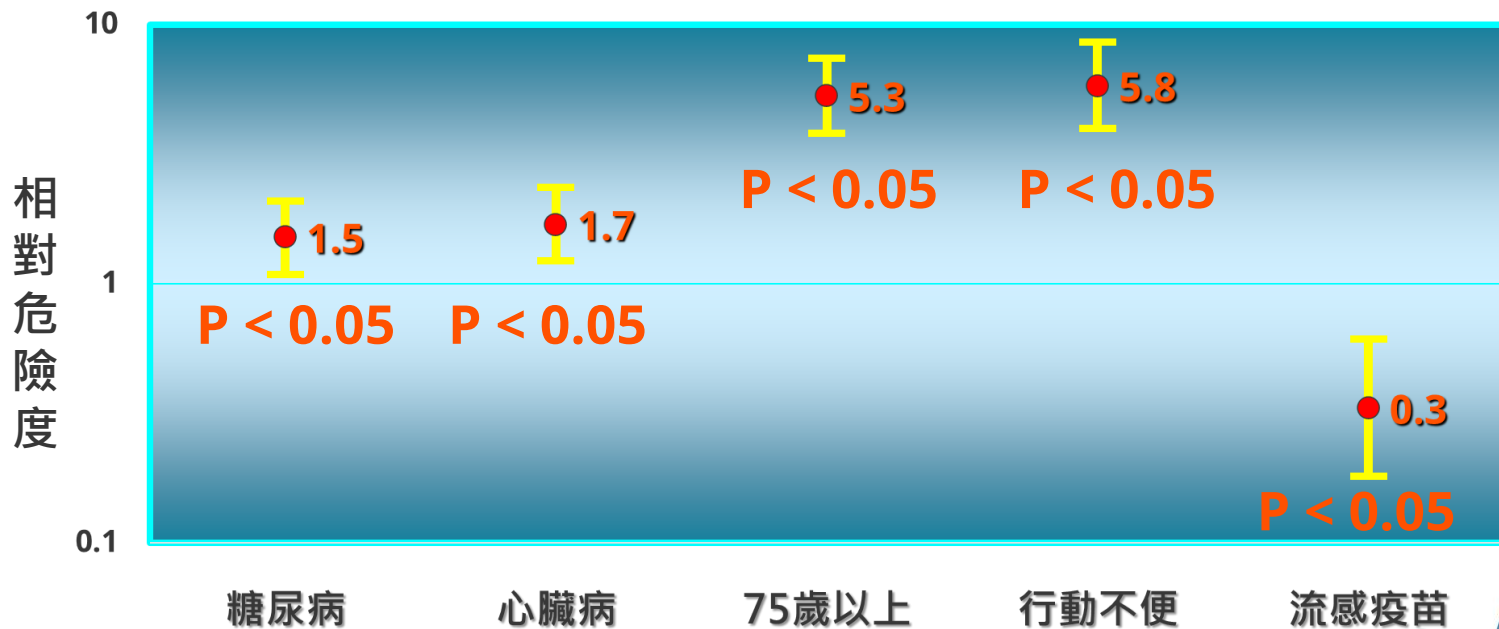
橘子新聞 20XX年X月X日

- 外電報導，最近以色列針對55歲以上老人，研究他們於2003年至2007年接種流感疫苗之後的資料。**驚訝地發現**在接種疫苗後兩週內，每年有25至40名老人死亡.....
- 國內專家XXX表示，**不排除**流感疫苗可能提高老人的死亡率，**並不諱言**應該重新討論國內疫苗政策.....



流感疫苗可能致死？

背景：2006年10月四位以色列老人猝死
2007 以色列



資料來源：
Kokia ES. Vaccine 2007;25:8557.

Recommended composition of influenza virus vaccines for use in the 2024-2025 northern hemisphere influenza season

WHO

■ Trivalent vaccines:

➤ Egg-based vaccines

- an A/Victoria/4897/2022 (H1N1)pdm09-like virus;
- an A/Thailand/8/2022 (H3N2)-like virus; and
- a B/Austria/1359417/2021 (B/Victoria lineage)-like virus.

➤ Cell culture- or recombinant-based vaccines

- an A/Wisconsin/67/2022 (H1N1)pdm09-like virus;
- an A/Massachusetts/18/2022 (H3N2)-like virus; and
- a B/Austria/1359417/2021 (B/Victoria lineage)-like virus.

Recommended composition of influenza virus vaccines for use in the 2024-2025 northern hemisphere influenza season

WHO

- Quadrivalent vaccines : While the **B/Yamagata lineage vaccine component should be excluded** as it is no longer warranted, where quadrivalent vaccines are still in use, the B/Yamagata lineage component remains unchanged:
 - a B/Phuket/3073/2013 (B/Yamagata lineage)-like virus.